Virtual Bench Pro4.0 User Manual

Version 1.2

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1 Environment setup

1.1 Hardware

To use Virtual Bench Pro 4.0 (VBP4.0), connect PC with EVB board and USB to I2C Communication Device (EVKT_USBI2C_02) are necessary.

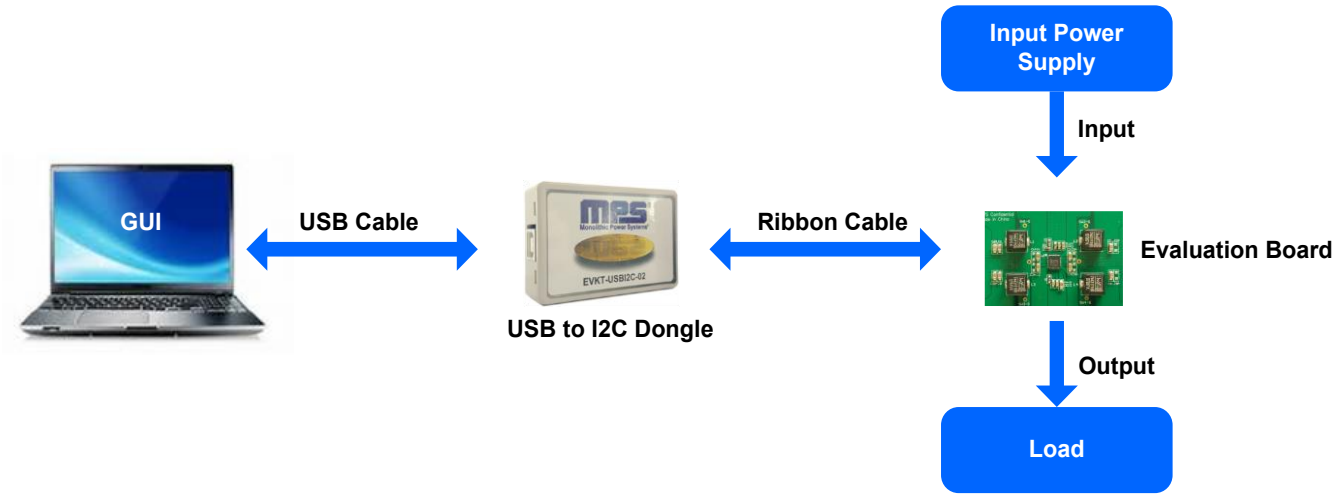


Figure 1. Evaluation Board Setup

1.2 System requirement

#	Software/Hardware	Version
1	Operating System	Windows 7 or later
2	.Net Framework	.NET Framework 4.0 or later
3	MPS Dongle	R5 or later

1.3 Driver install

The I2C driver should be installed before using VBP4.0. The driver will be published with VBP4.0.

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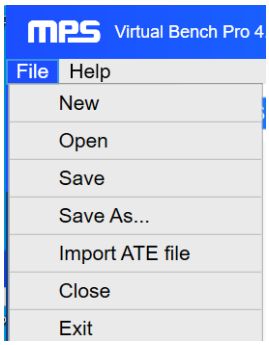
2 Introduction

VBP4.0 is a Graphical User Interface (GUI) supported by MPS used to configure and monitor multiple MPS digital power solutions with an I2C Interface. This GUI is an intuitive tool that can be used with MPS evaluation kits and helps assist in the evaluation and in the testing phases of your design.

2.1 Menu Bar

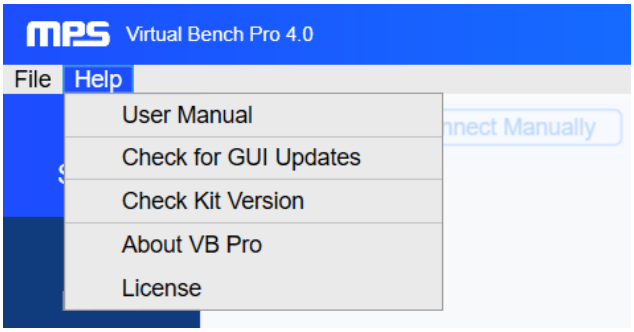
The main menu of VBP4.0 provides convenient access to many important functions, like open and save project files, create new project, check kit firmware version, license control, etc. See Main Menu for more information.

1) File



- **New**: Create a new project.
- **Open**: Open an existing project.
- **Save**: Save the current project.
- **Save As…**: Save the current project with different file name.
- **Import ATE file**: import configurations from an existing ATE file.
- **Close**: Close the current project.
- **Exit**: Exit VBP4.0.

2) Help



- **User Manual**: Display User Manual for VBP4.0.
- **Check for GUI Updates**: Check if there is a newer version of VBP4.0.

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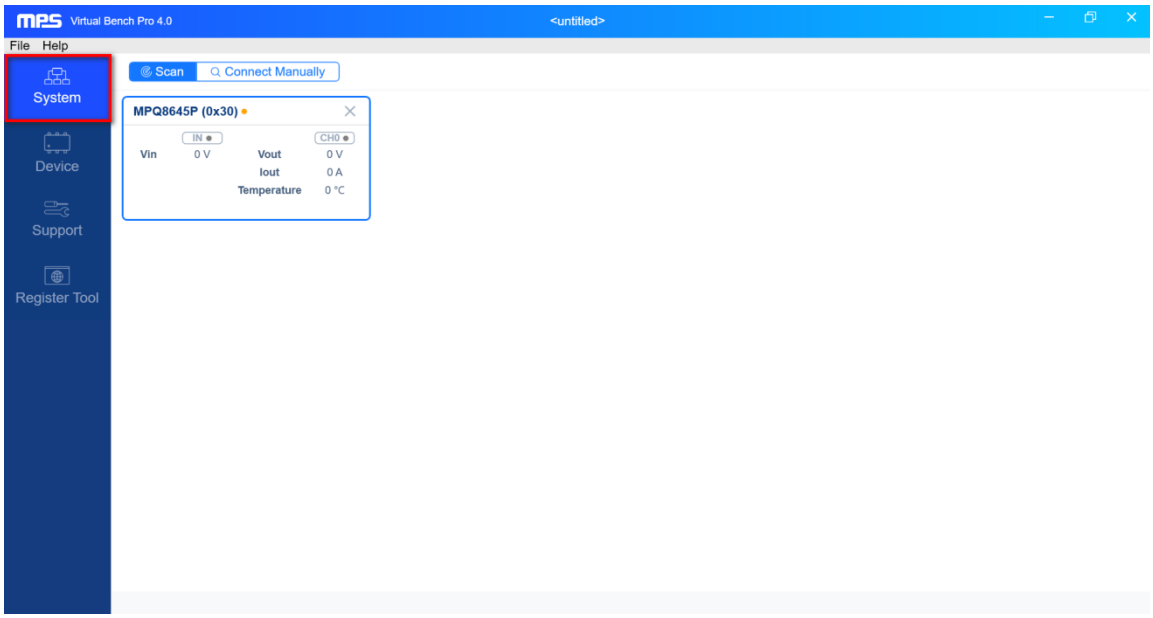
- **Check Kit Version**: Check for the dongle version and support to update dongle firmware to latest release version when the firmware version is outdated. Please note that the update function is only available for EVKT-USBI2C-02 and EVKT-AUTOMOTIVE-00A. Also, some older dongles may not display the version number, and update function is not supported.
- **About VB Pro**: Display VBP4.0 version and copyright.
- **License**: Interface to enter and display VBP4.0 license information.

2.2 Navigation Bar

Navigation Bar lets user navigate between different function panels.

1) System

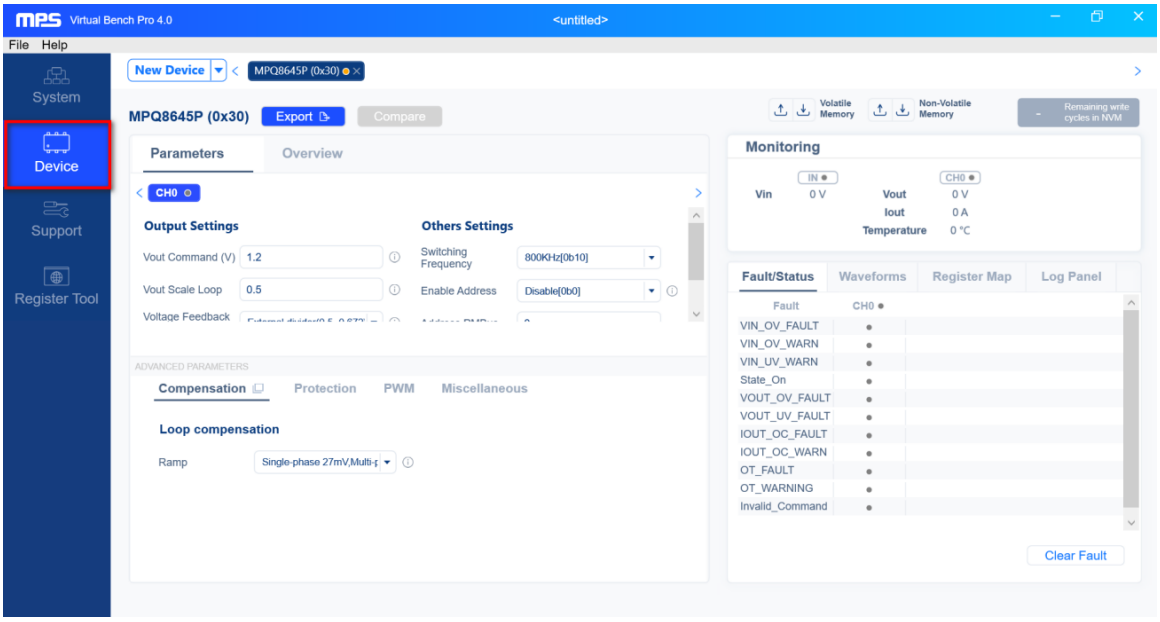
System Panel allows users to add/remove chips and displays an overview of the chip information, like connection status, part numbers, and monitor values.



2) Device

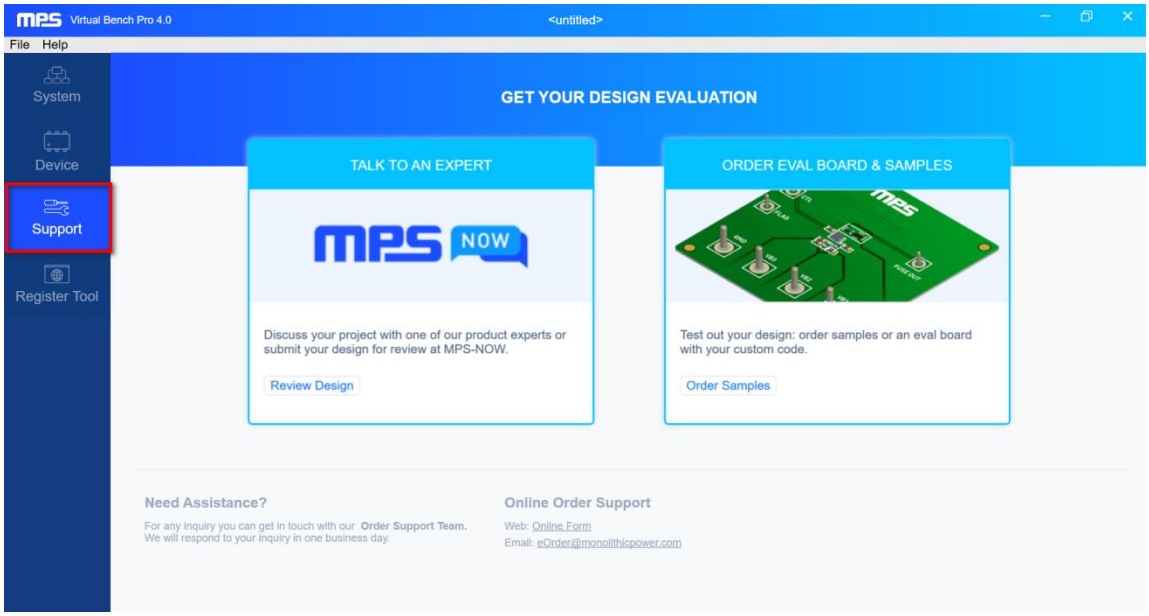
Device panel displays detail information of the chip and allows user to configure the chip.

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3) Support

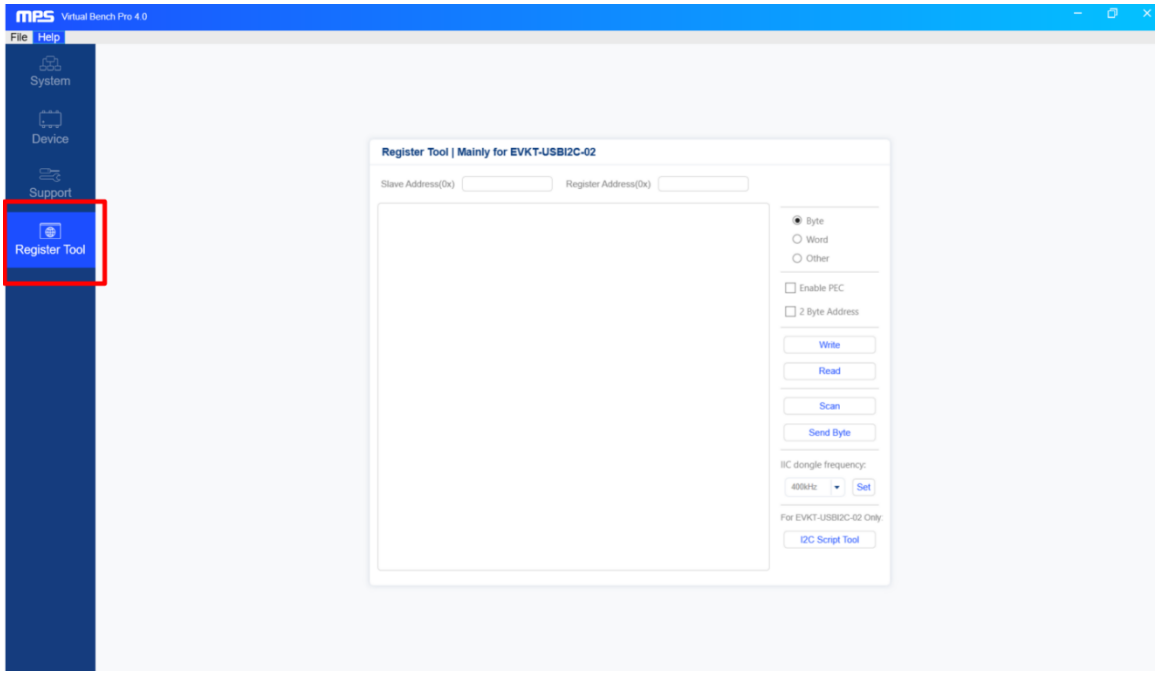
Support panel provides support links to MPS sites.



4) Register Tool

Register Tool Panel provides additional operations to the connected part.

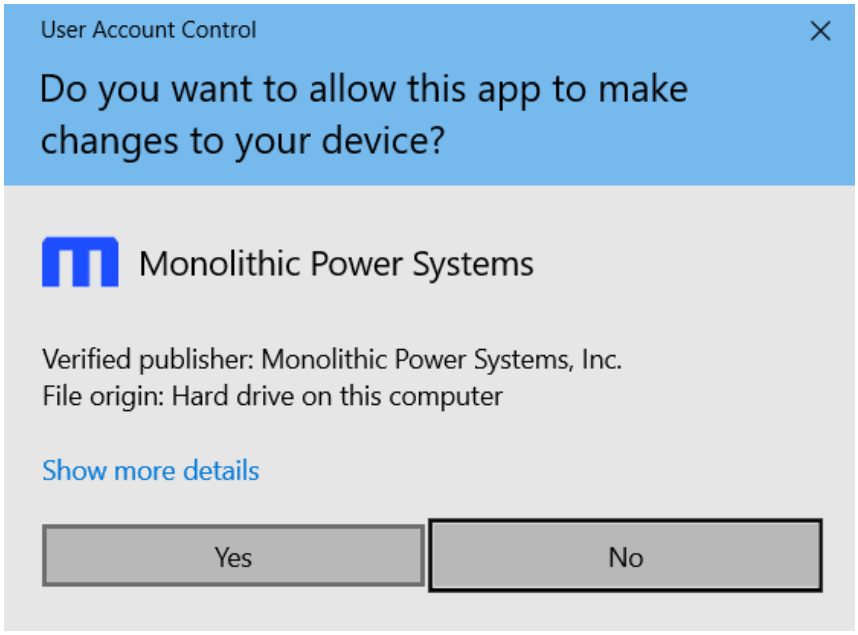
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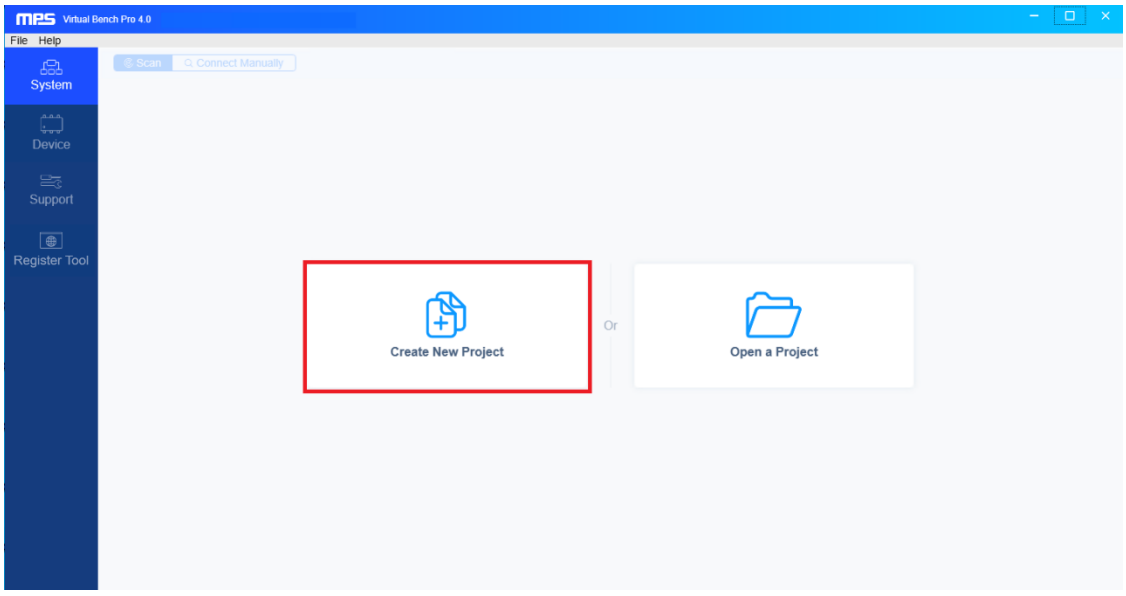
3 How to use

Make sure the hardware setup is good, then launch VBP4.0. Windows system may display **User Account Control**, click **Yes** to continue.



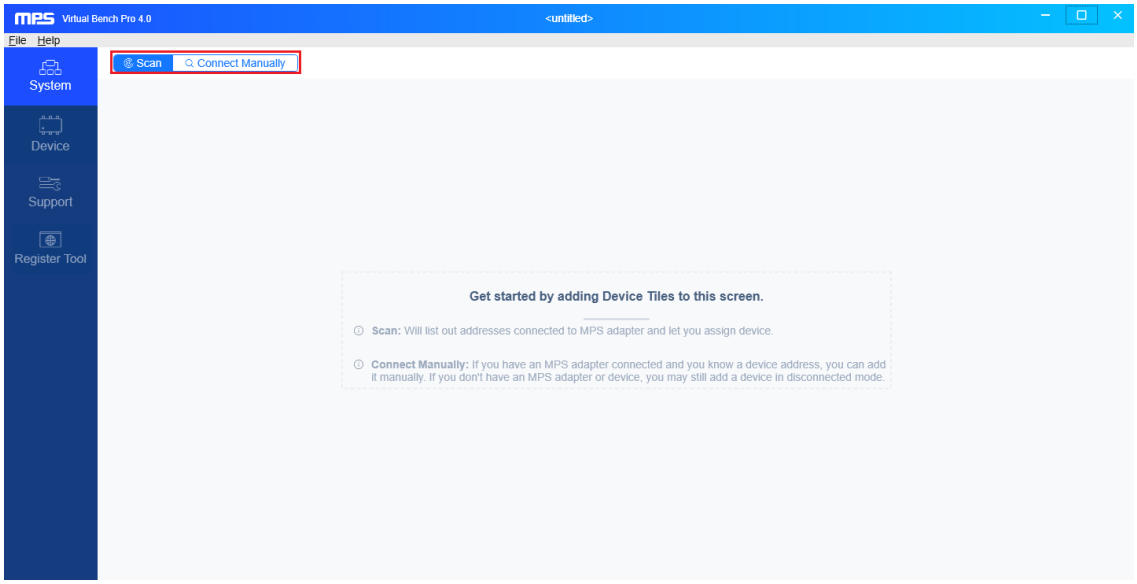
3.1 Add New Device

Click **Create New Project**.



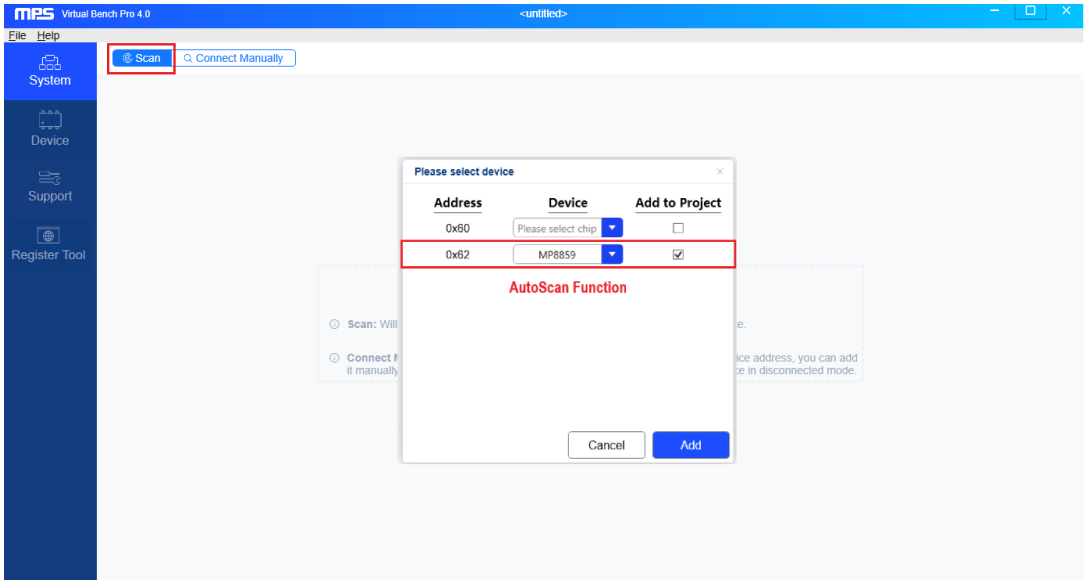
Two ways are provided by VBP4.0 to add a new device: Scan method and Connect Manually method.

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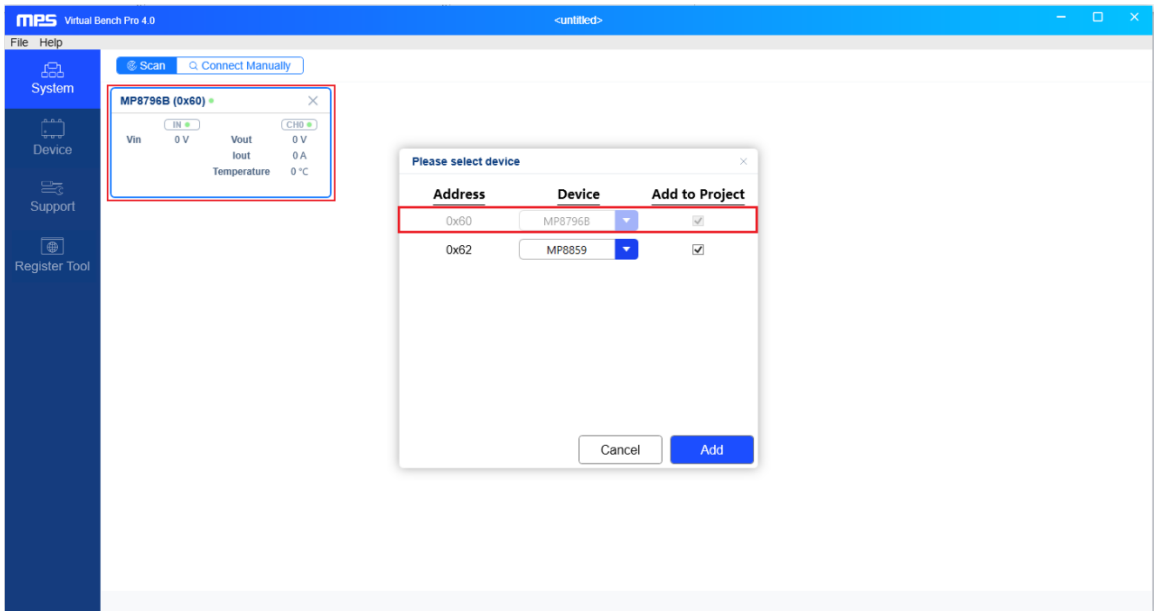
1) Scan Device

Scan method, if a part supports auto scan function, when click Scan button, this part will be detected automatically; Otherwise, when click **Scan** button, just the address can be found, user should select the part manually.

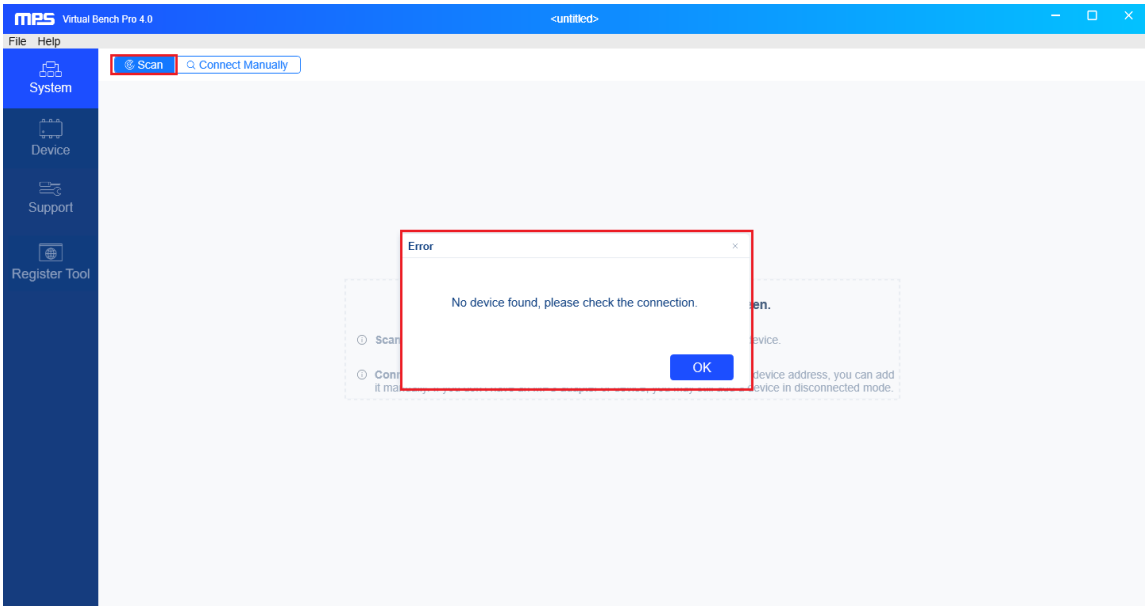


Note: if a part with address already in VBP4.0, the same address found by **Scan** function can't be added to VBP4.0 again.

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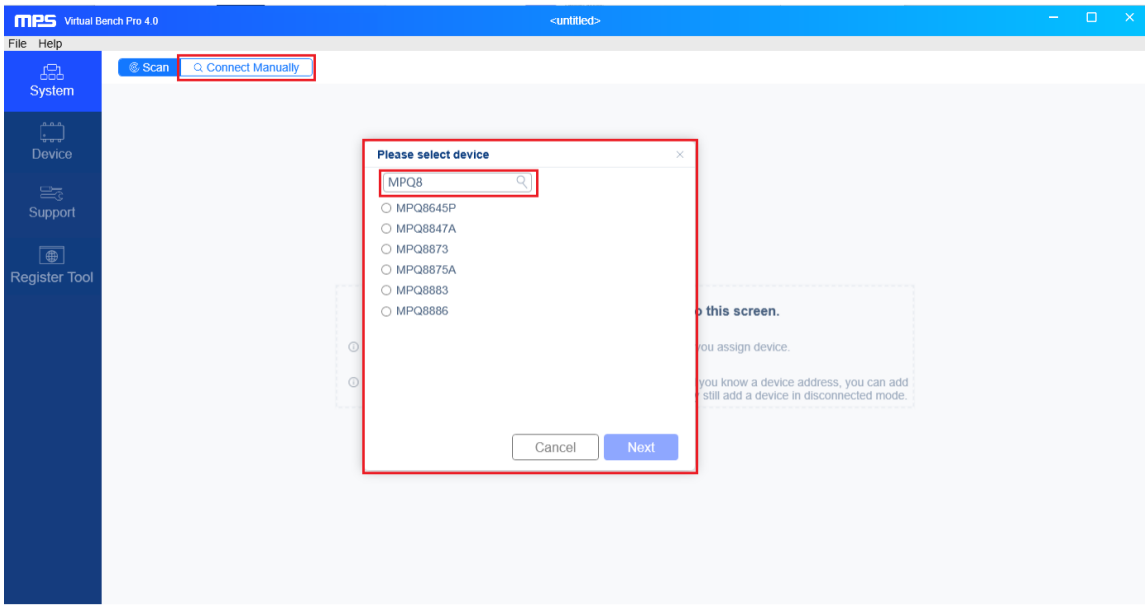
If there is no part found by **Scan** function, Error window with “No device found, please check the connection” will pops up. Please check the connection.



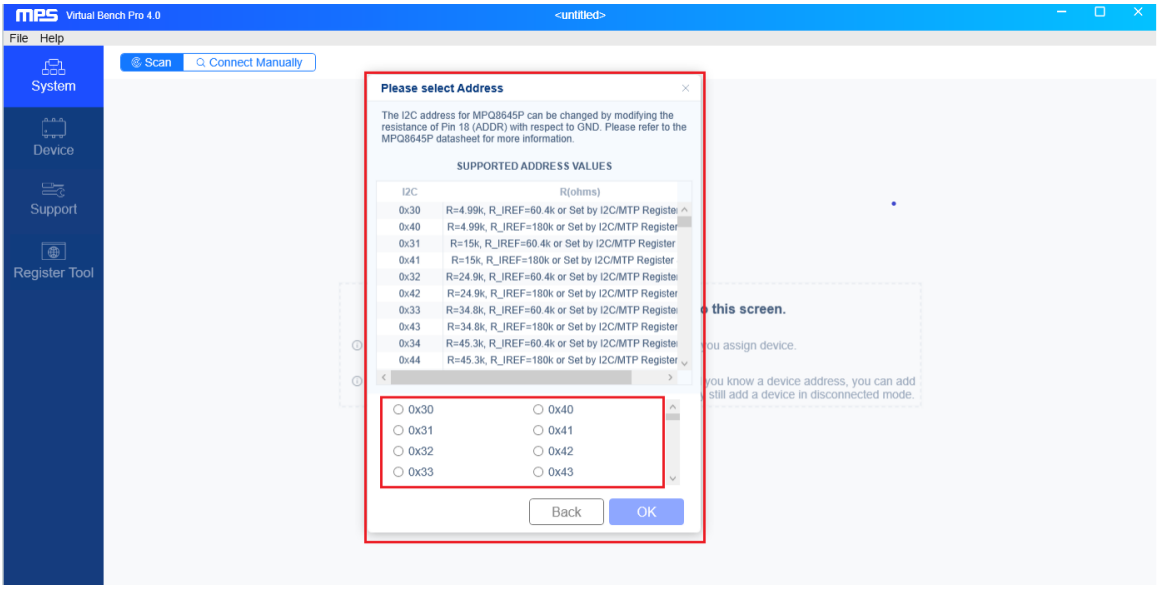
2) Connect Manually

Connect Manually method, this provides a method for customers to add a device to VBP4.0 manually. Click **Connect Manually** button, device select window which displays all the supported devices will pop up. A search function is supported for users to filter desired devices in this window. This function supports fuzzy search and is case-insensitive for chip names. Choose the correct device, and then click **Next** button.

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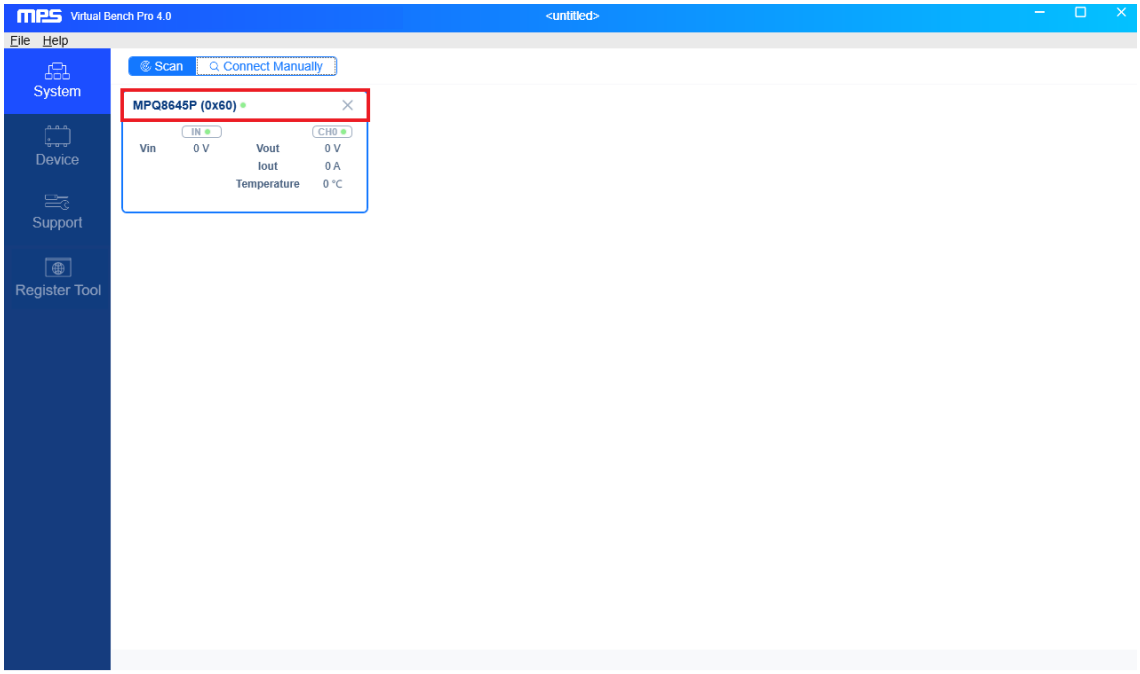
A sub window with the I2C addresses recommended by MPS will pop up, choose the correct Address from the list, and then click **OK** button. The chosen device will be added.



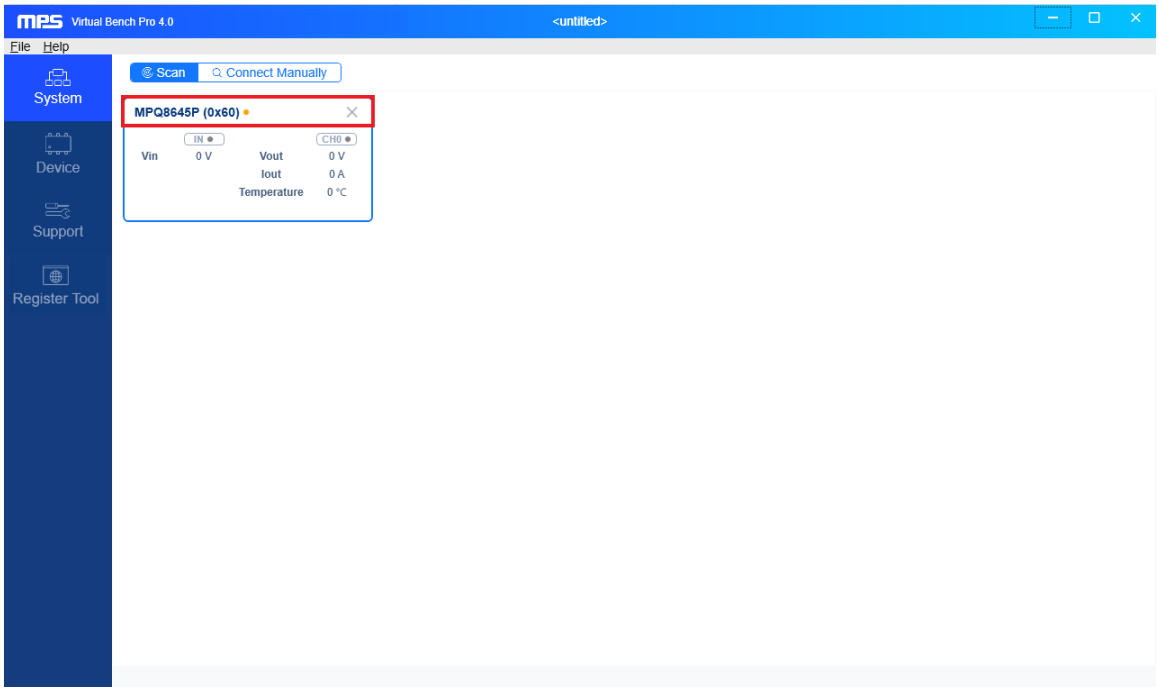
3.2 Device Presetting

Now a new device has been added. If the light indicator behind the part number is green, the connection is correctly established.

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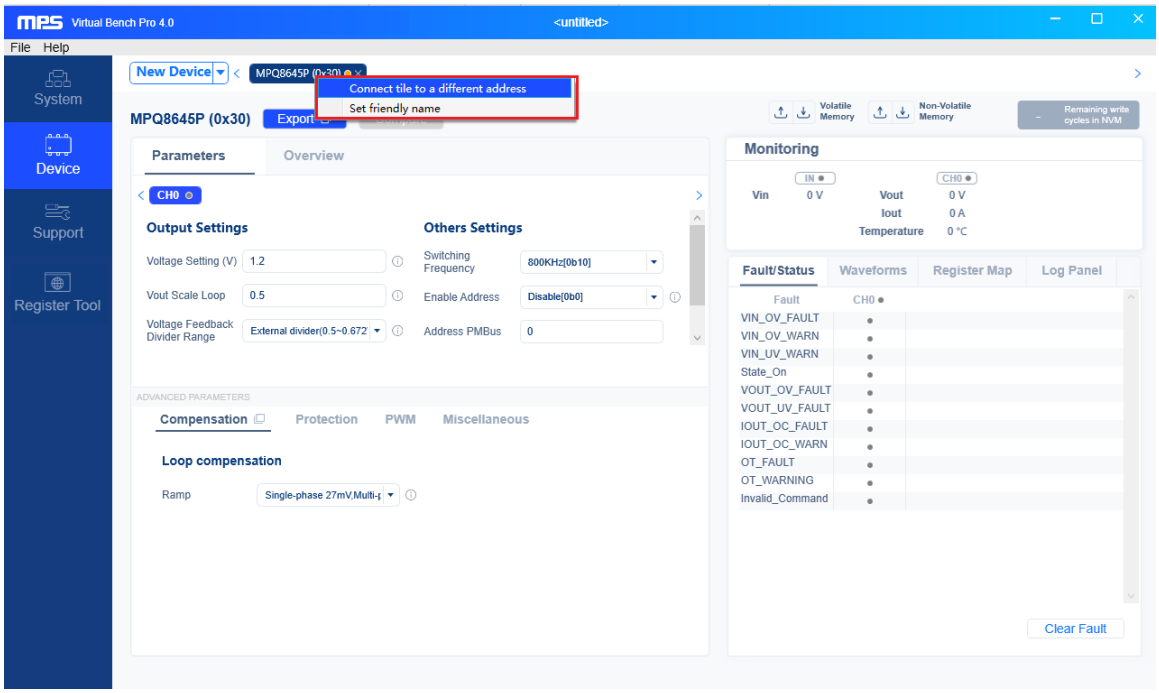


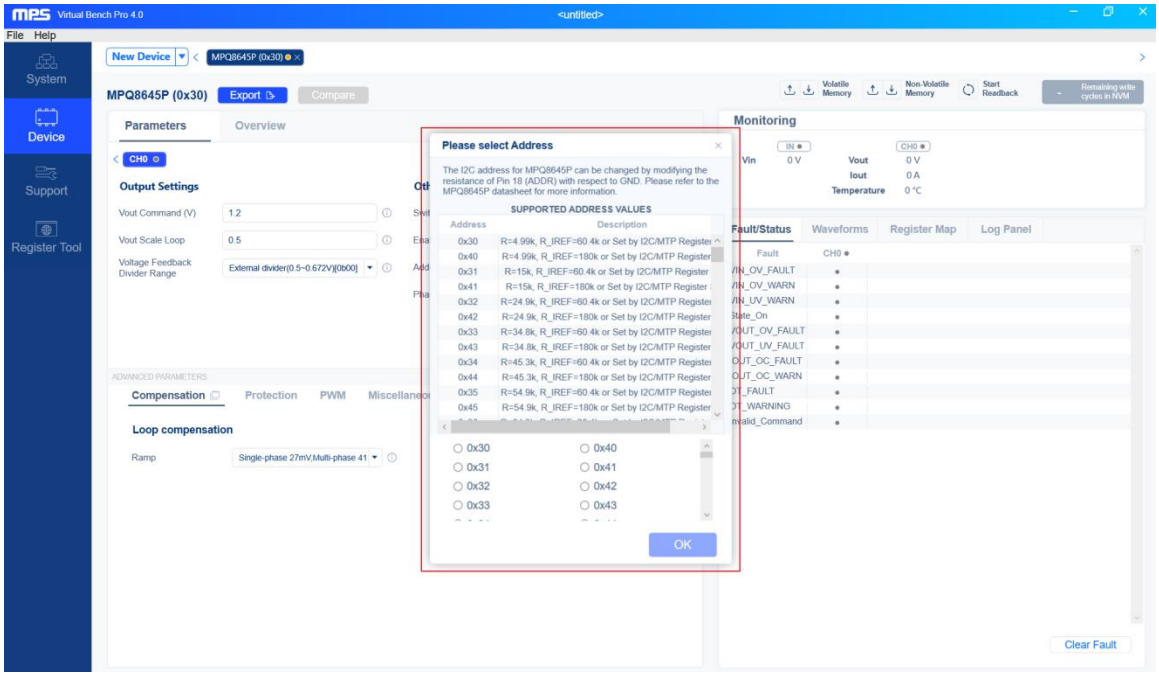
If the light indicator behind the part number is not green, there is connection issue.



User can right click the device in Device panel, select “Connect tile to a different address”, the user will be allowed to re-select the device address.

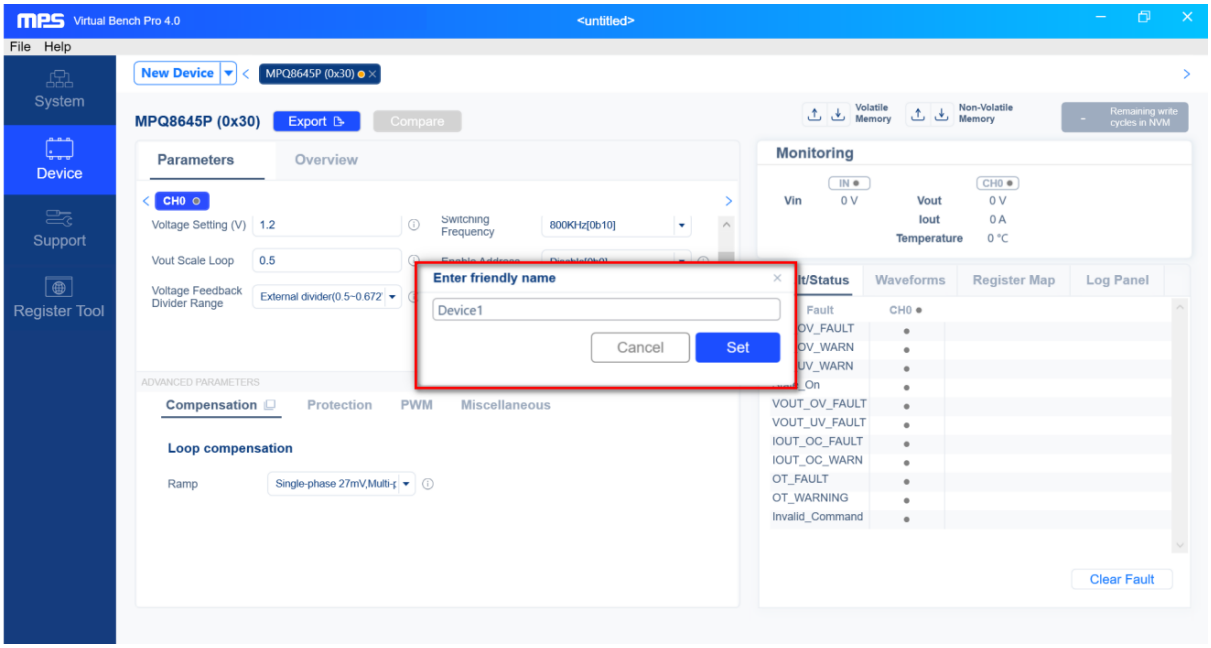
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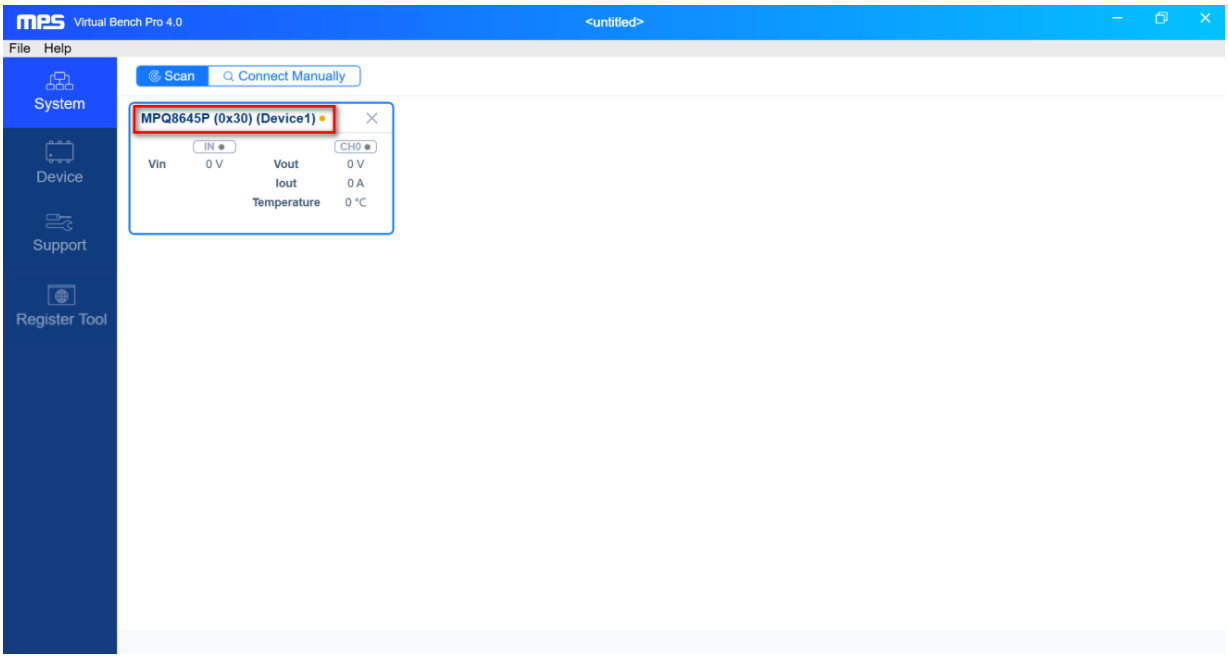


Once “Set friendly name” is clicked, a window will pop up so that user can enter the desired friendly name to be set.

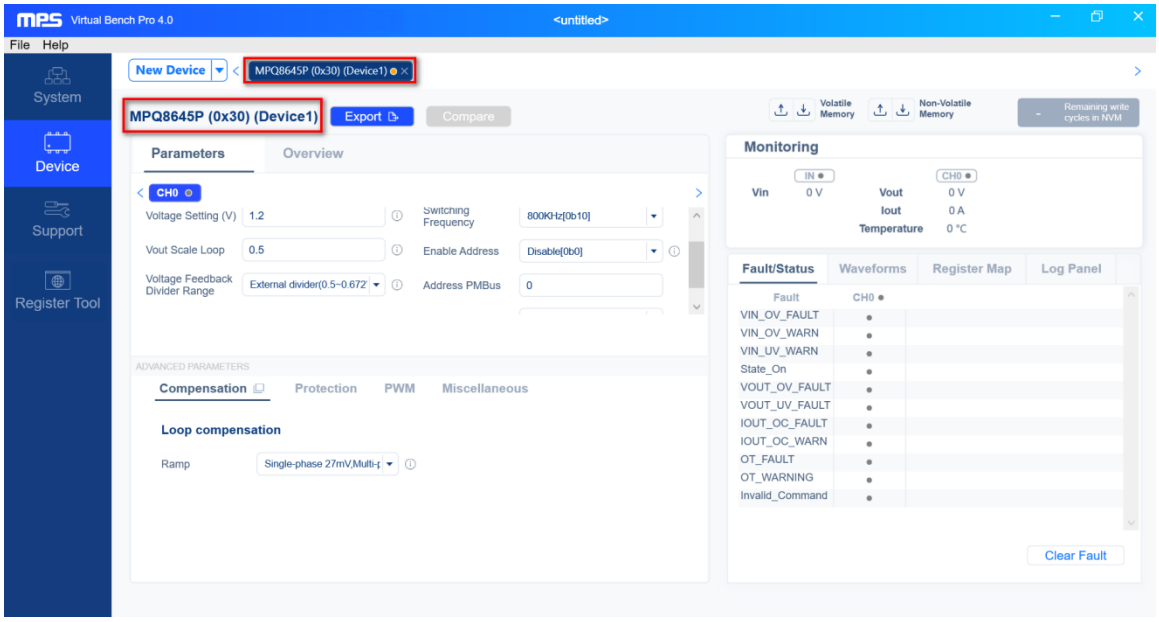
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After **Set** Button is clicked, the Friendly Name will be set and displayed in both System and Device Panel.

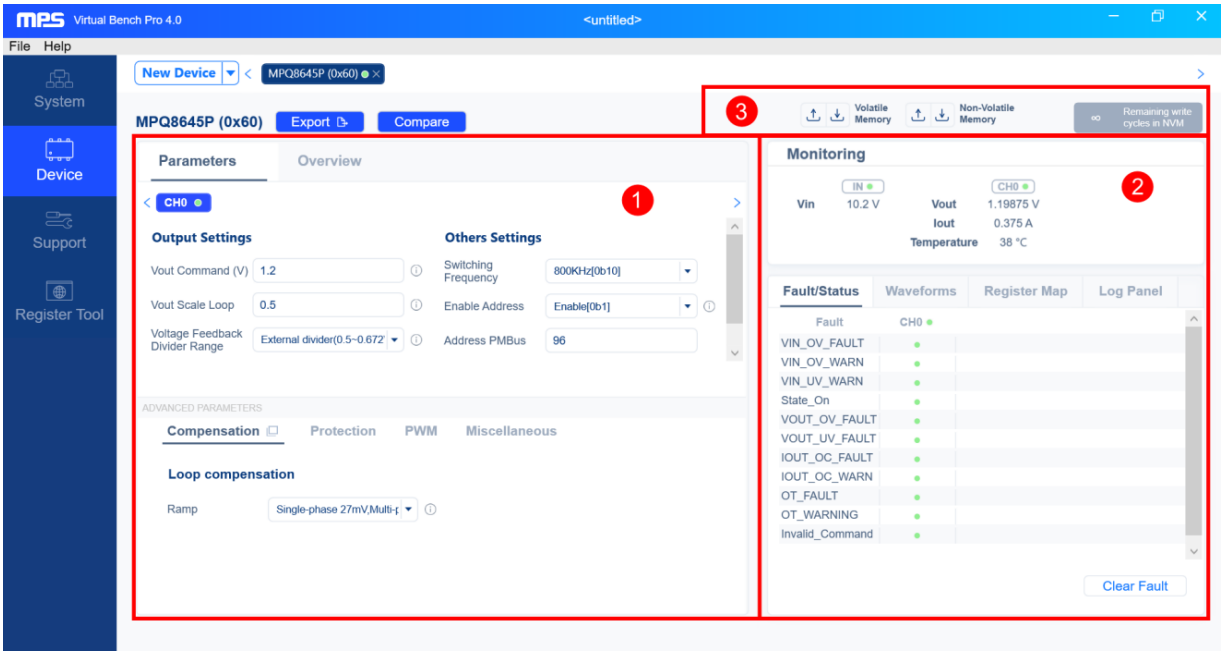


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3.3 Device Configuration

Assume a correct connection is established, user can control this chip in the device panel. All the writable control bits in Data Sheet can be found in **PART1**. And all the monitor values, fault information, operation logs can be found in **PART2**. All the updated values in part1 can be written to chip on-line via buttons in **PART3**.

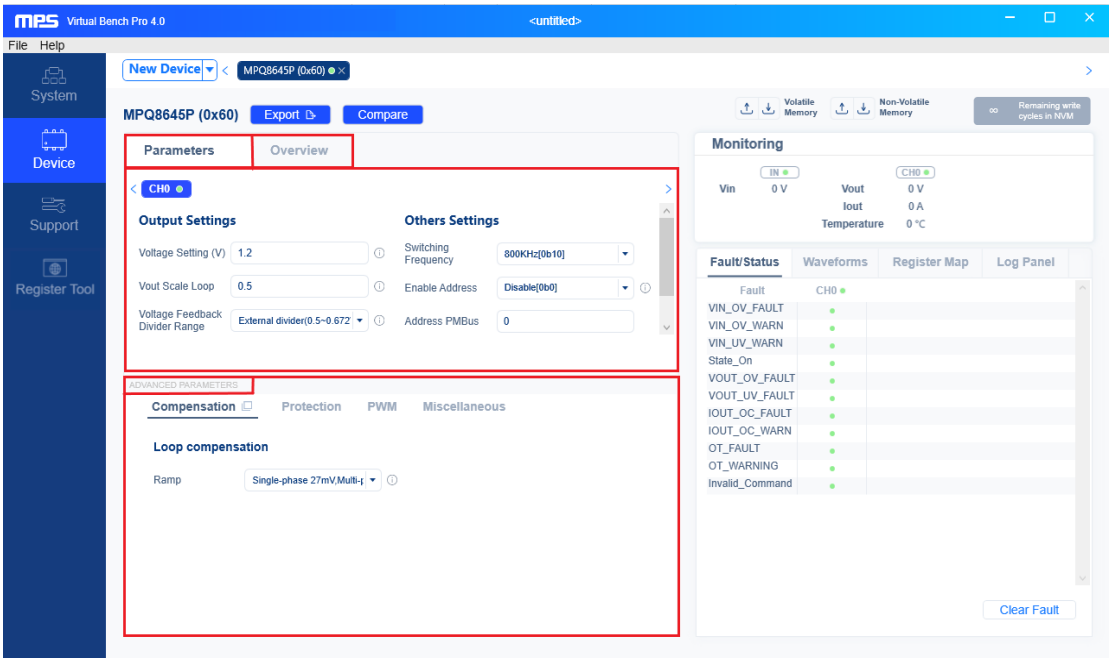


1) PART1

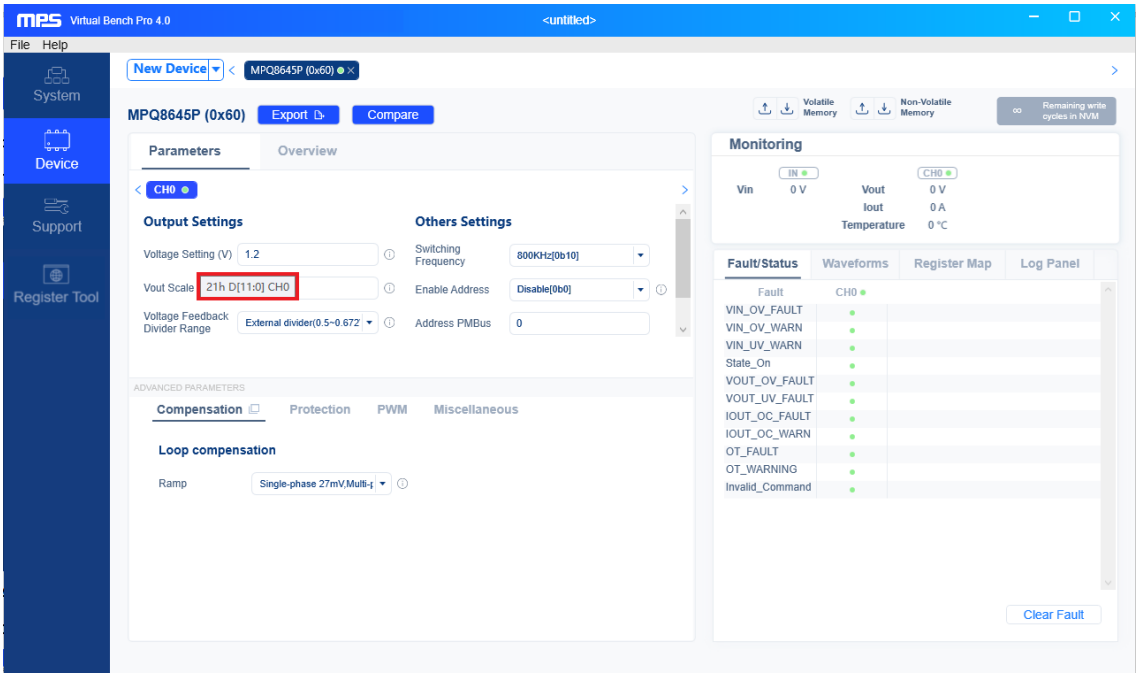
There are three portions in part1, **Parameters**, **Overview**, **Advanced Parameters**. Users can see the basic configurations of the chip in **Parameters**, and description of the chip in **Overview.** All other

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settings can be found in **Advanced Parameters**, and the settings are divided into different groups, like **Compensation**, **PWM**, etc.

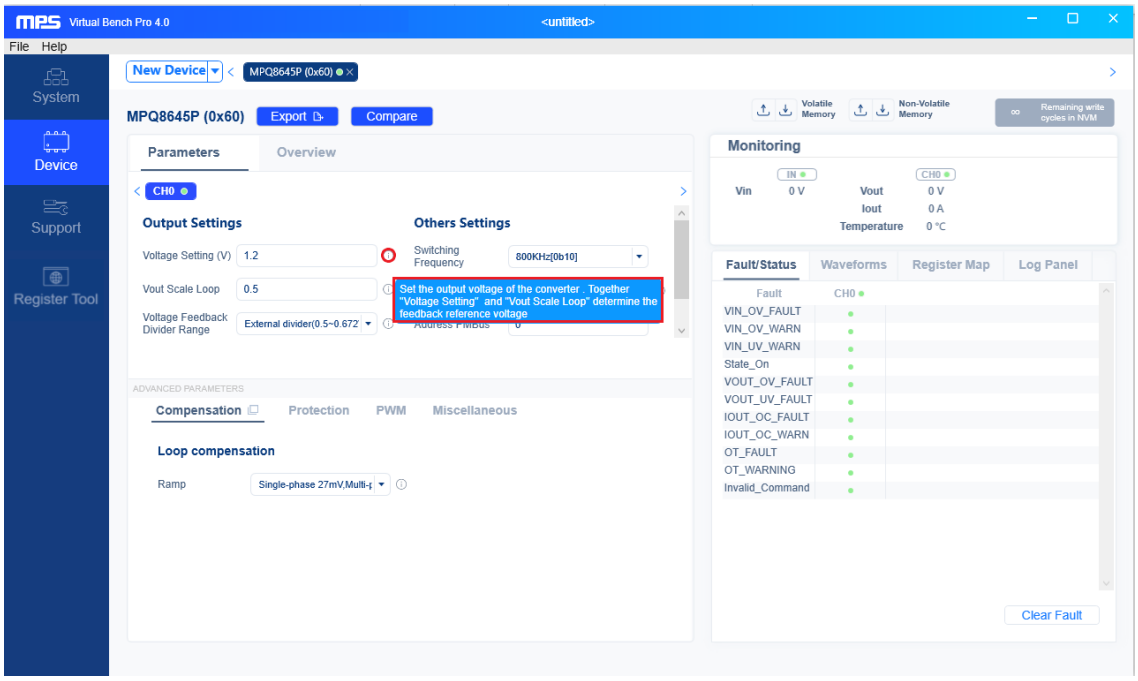


Parameter's tip will pop out when hovering mouse over the parameter's name, including register and page information.

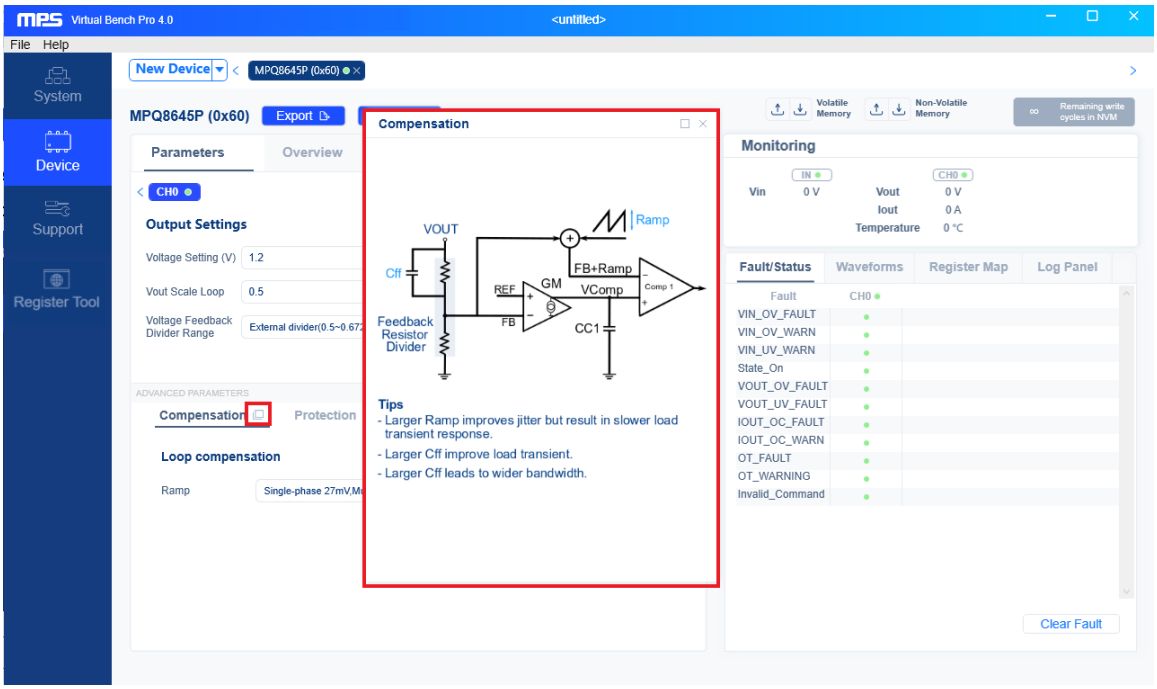


A prompt window will pop out when hovering over the icon ⓘ at the end of the parameter, recording description of this parameter.

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A prompt window will pop up when clicking on the icon ▢ in the **Advanced Parameters** part.

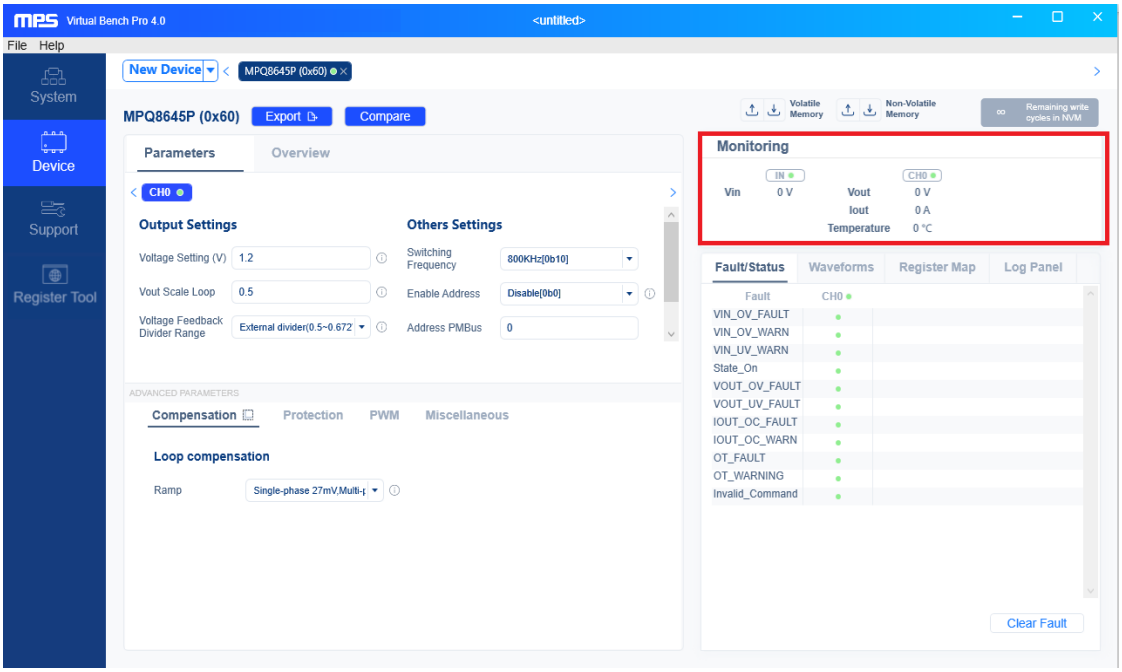


2) PART2

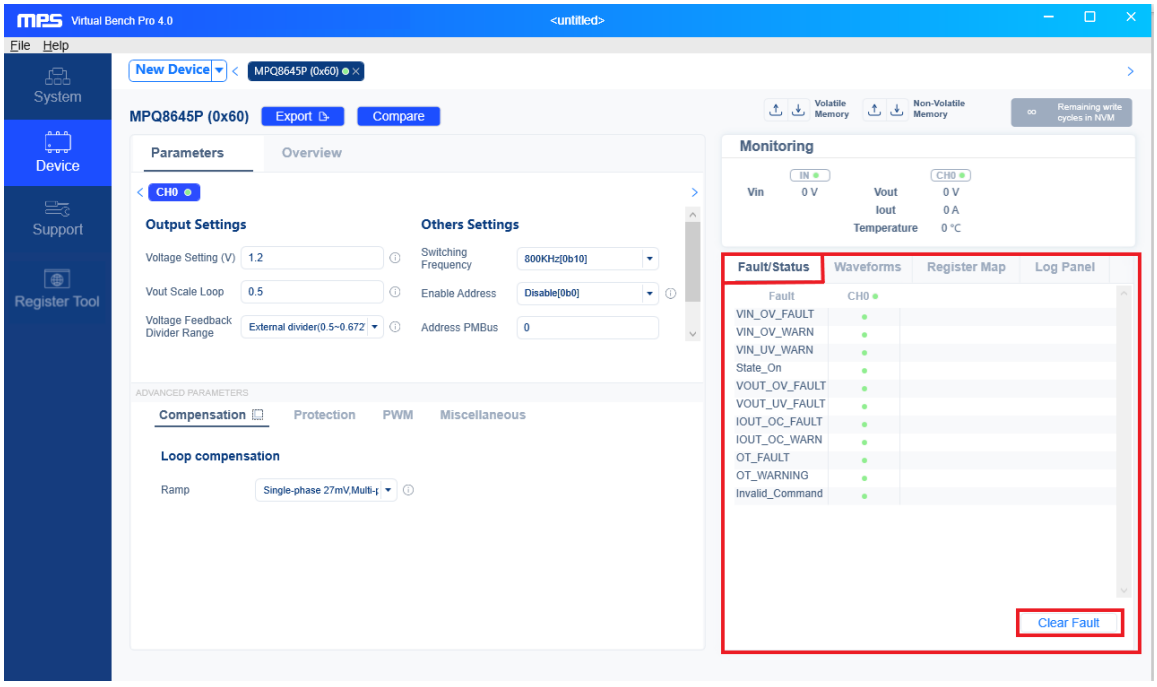
There are **Monitoring**, **Fault/Status**, **Register Map**, **Waveforms**, and **Log Panel** sections.

- **Monitoring**: Shows the monitor values of the chip in real time.

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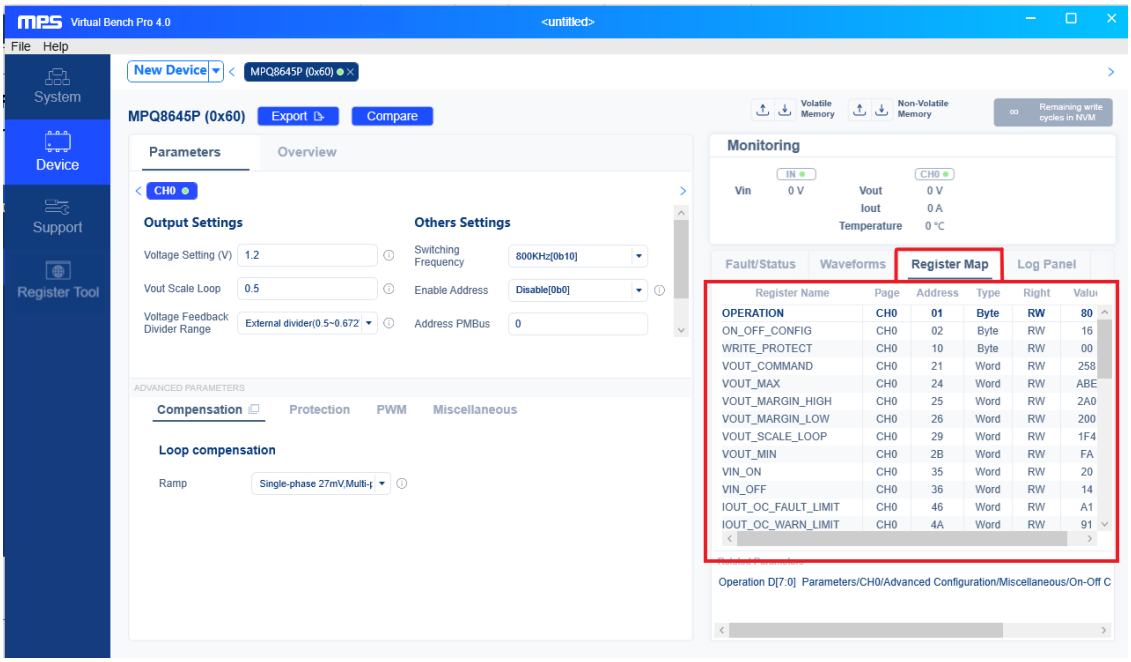


- **Fault/Status**: Displays the fault and status information. The green light indicates there is no fault of the chip, while the red light indicates some faults are triggered, and users can click **Clear Fault** button to clear those faults.

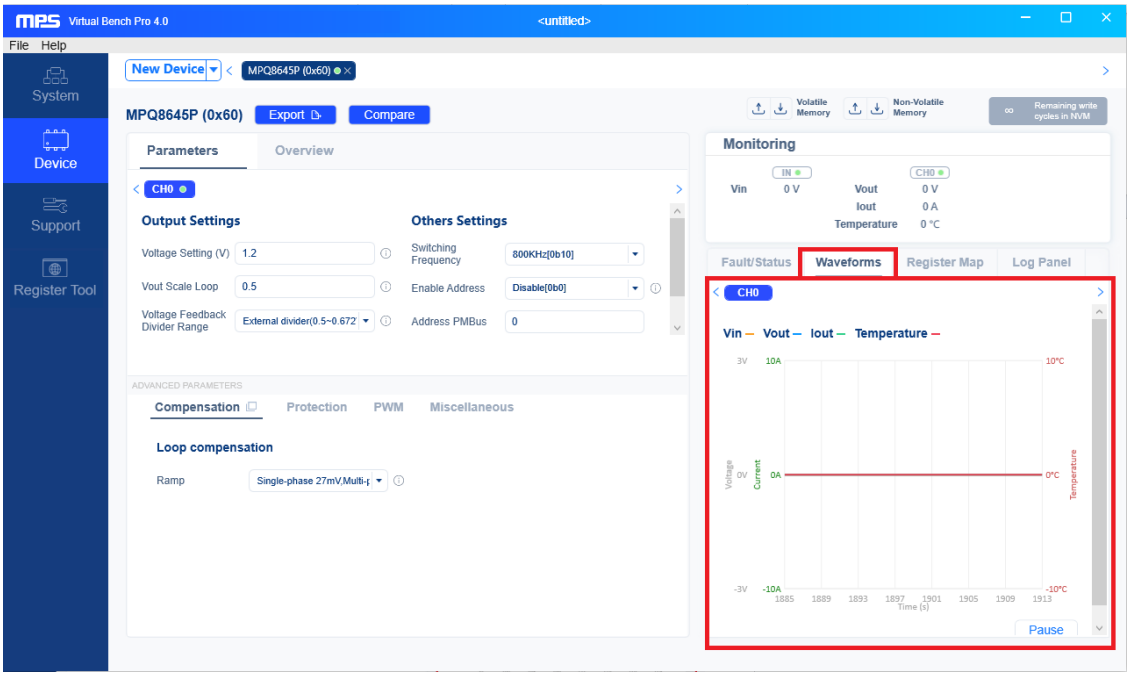


- **Register Map**: Shows all the register values of the chip but the tab doesn't display the register values in real time. User needs to click **Read RAM** button to get register values in real time.

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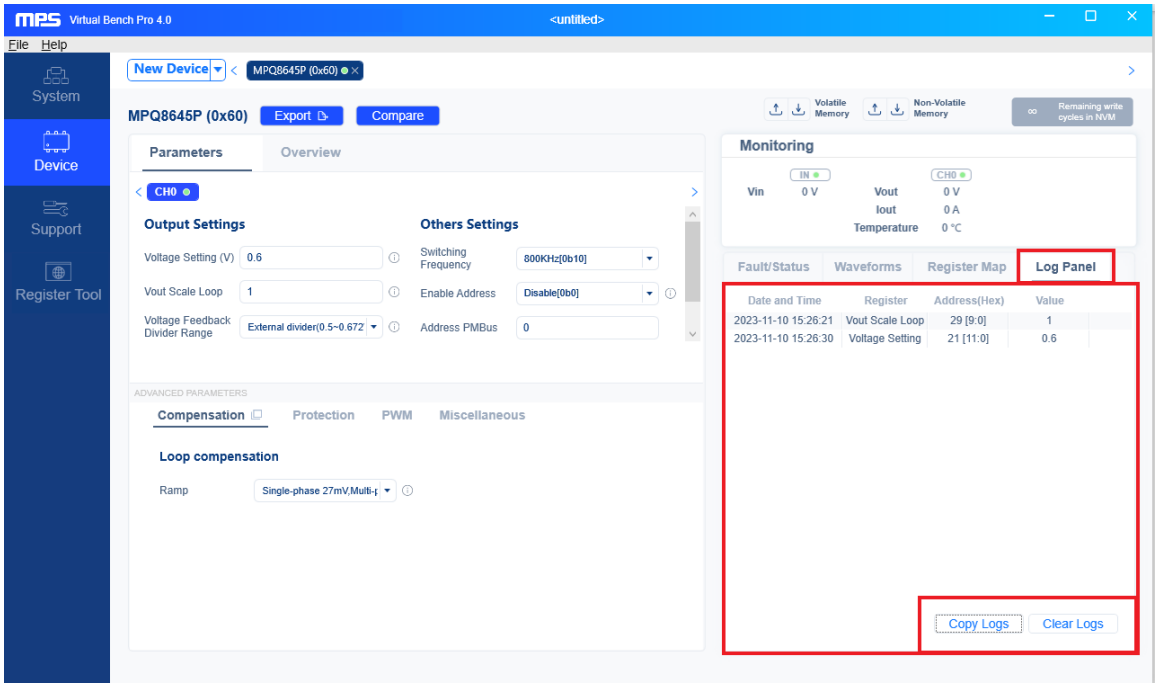


- **Waveforms**: Shows continuous monitoring value for 1 minutes.



- **Log Panel**: Record all the operations user perform on the chip. User can copy logs by clicking **Copy Logs** button, and clear logs by clicking **Clear Logs** button.

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3) PART3

Here, users can select read/write operations. **Volatile Memory** and **Non-Volatile Memory** operations can be found in this panel.

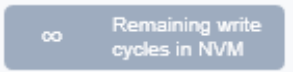
Volatile Memory operation includes Write/Read RAM, it provides a method to change chip value on-line, if the power supply is shut down, the value will be returned to default value.

Non-Volatile Memory operation includes Write/Read ROM, it provides a method to save the volatile memory to the chip permanently, even the power is shut down. (Note: not all the chips support Non-Volatile Memory operation).

Remaining write cycles in NVM indicates how many times remain for Non-Volatile operation:

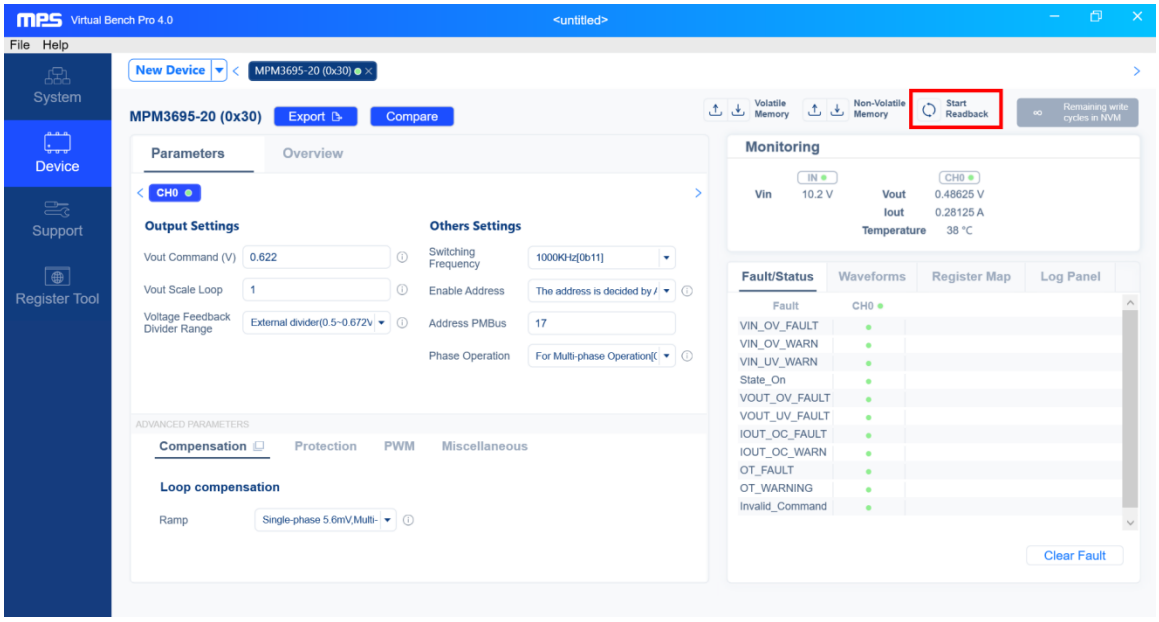
- "-" indicates this chip does not support **Non-Volatile operation**.
- "∞" indicates this chip supports **Non-Volatile operation** and has unlimited MTP times.



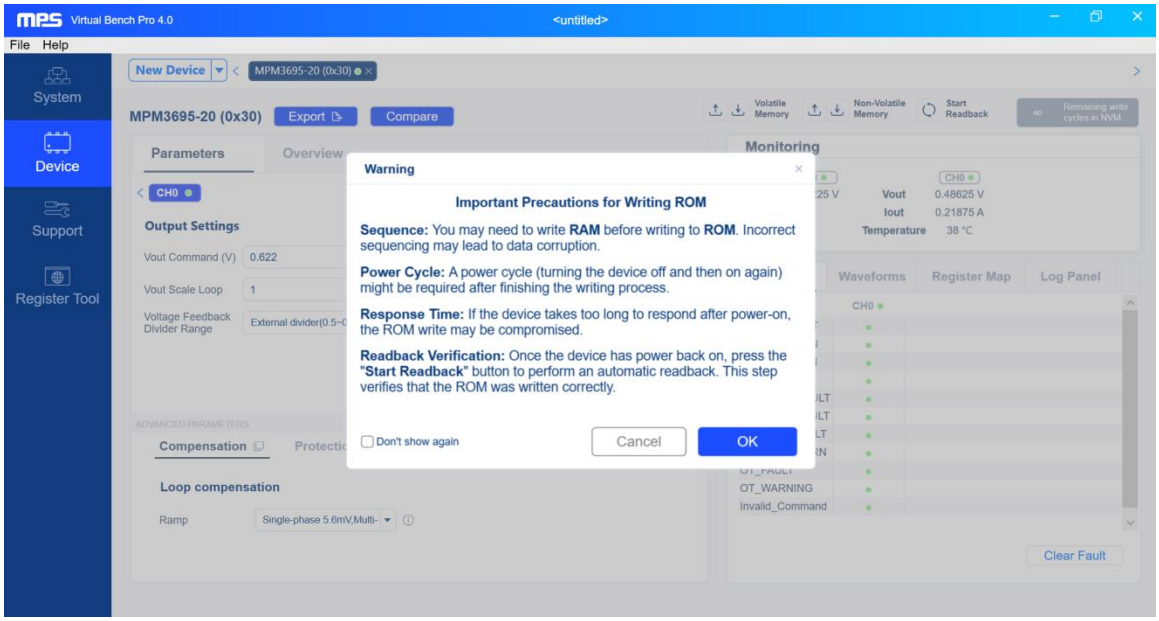


Start Readback is supported by some parts to verify the ROM was written correctly.

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Before proceeding, ensure that you have followed the correct sequence for writing the ROM. The tips will be shown by a pop-up Window as below after clicking “Write ROM”. The Window will not be shown next time if “Don’s show again” checked.

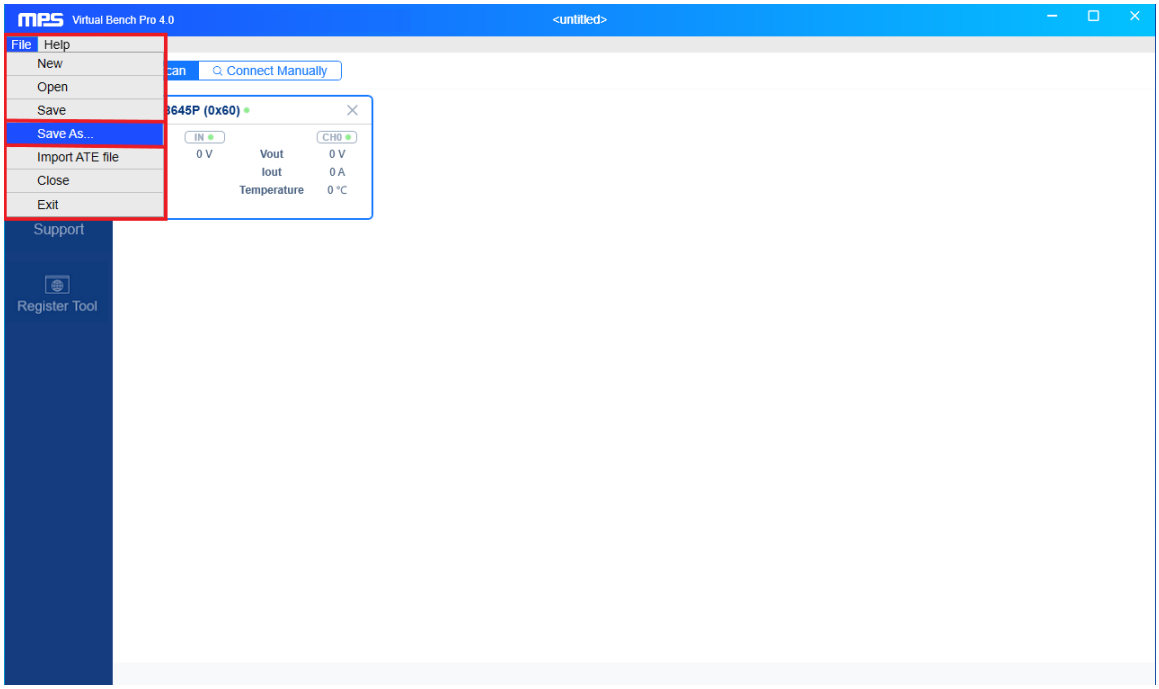


3.4 Save Configuration to Local

1) Save to Spec File

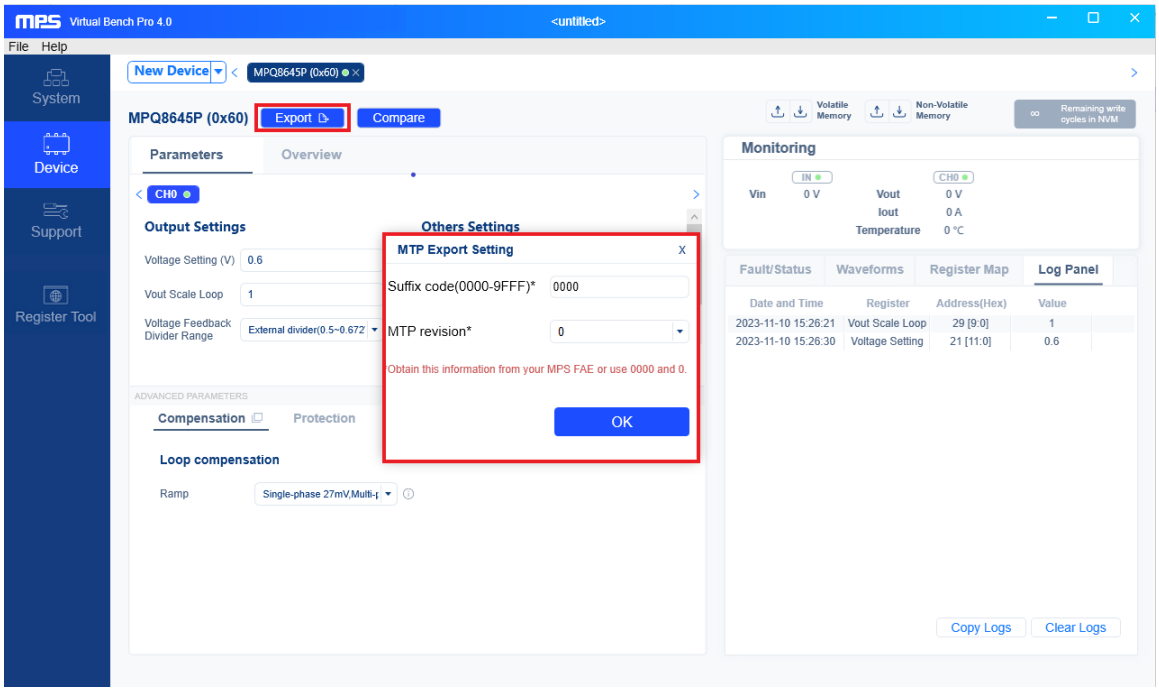
Users can save configurations for the next usage. Select **File->Save as** to save configurations or select **File->Open** to load the saved configurations.

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2) Save to ATE txt File:

In the Device panel, users are also allowed to save digital code for this configuration and send this code to MPS for further support. Select **Export** and record **Suffix code** and **MTP reversion** to save configurations to digital code.



Click **OK** button and choose a directory path, all configurations will be saved in the text file.

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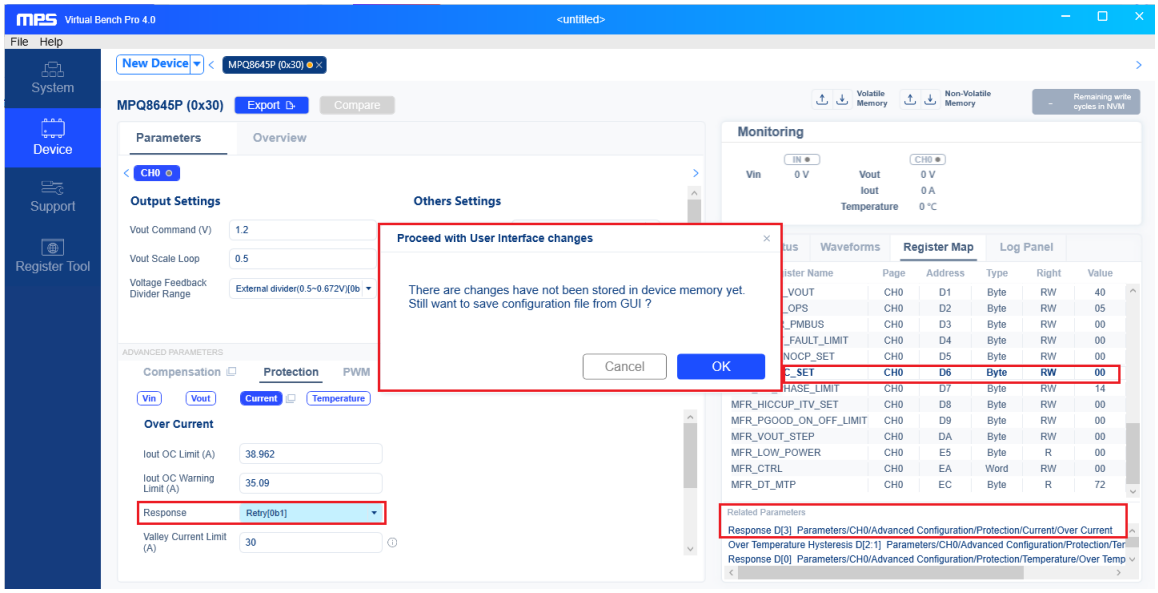
MPQ8645P-0000-rev0.txt - Notepad

File Edit Format View Help

```
0000    0       01      1       OPERATION       80      128
0000    0       02      2       ON_OFF_CONFIG   1E      30
0000    0       10      16      WRITE_PROTECT   00      0
0000    0       21      33      VOUT_COMMAND    0384    900
0000    0       24      36      VOUT_MAX        0BB8    3000
0000    0       25      37      VOUT_MARGIN_HIGH        03F0    1008
0000    0       26      38      VOUT_MARGIN_LOW 02FF    767
0000    0       29      41      VOUT_SCALE_LOOP 00FA    250
0000    0       2B      43      VOUT_MIN        00FA    250
0000    0       35      53      VIN_ON  0020    32
0000    0       36      54      VIN_OFF 0014    20
0000    0       46      70      IOUT_OC_FAULT_LIMIT     00A1    161
0000    0       4A      74      IOUT_OC_WARN_LIMIT      0091    145
0000    0       4F      79      OT_FAULT_LIMIT  009B    155
0000    0       51      81      OT_WARN_LIMIT   0091    145
0000    0       55      85      VIN_OV_FAULT_LIMIT      0021    33
0000    0       57      87      VIN_OV_WARN_LIMIT       0021    33
0000    0       58      88      VIN_UV_WARN_LIMIT       0010    16
0000    0       60      96      TON_DELAY       0000    0
0000    0       61      97      TON_RISE        0001    1
0000    0       64      100     TOFF_DELAY      0000    0
0000    0       D0      208     MFR_CTRL_COMP   01      1
0000    0       D1      209     MFR_CTRL_VOUT   40      64
0000    0       D2      210     MFR_CTRL_OPS    05      5
0000    0       D3      211     MFR_ADDR_PMBUS  00      0
0000    0       D4      212     MFR_VOUT_FAULT_LIMIT    00      0
0000    0       D5      213     MFR_OVP_NOCP_SET        00      0
0000    0       D6      214     MFR_OT_OC_SET   00      0
0000    0       D7      215     MFR_OC_PHASE_LIMIT      1A      26
0000    0       D8      216     MFR_HICCUP_ITV_SET      00      0
0000    0       D9      217     MFR_PGOOD_ON_OFF_LIMIT  00      0
0000    0       DA      218     MFR_VOUT_STEP   00      0
END
```

If there are some options changed on the GUI, a confirmation box will pop up. Click “OK”, modified values will be saved to txt file, otherwise, abandoned. The same function applies with **File->save/save as,** the modified values will be saved to spec file.

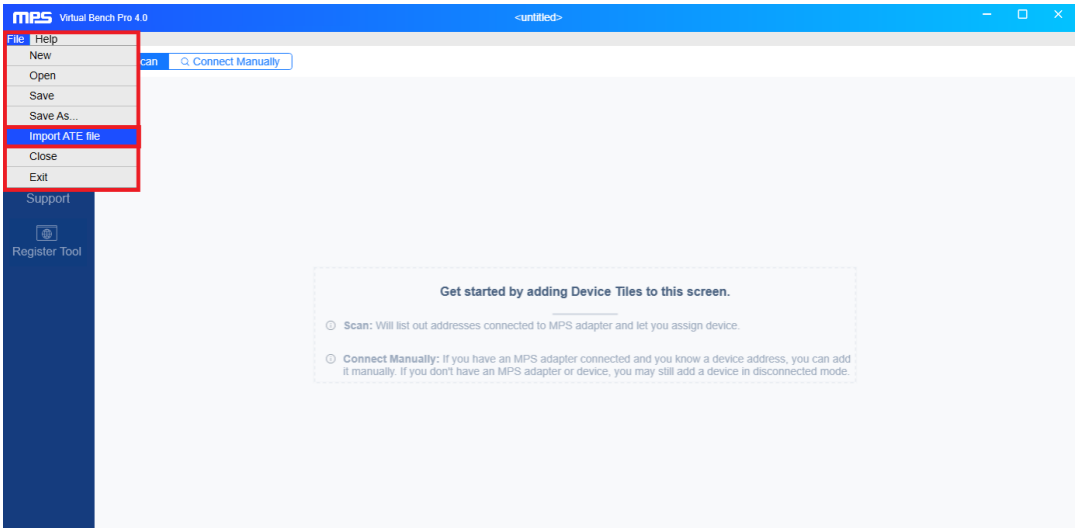
(Note: The modified values will not be written to the device, only be saved to txt/spec file.)



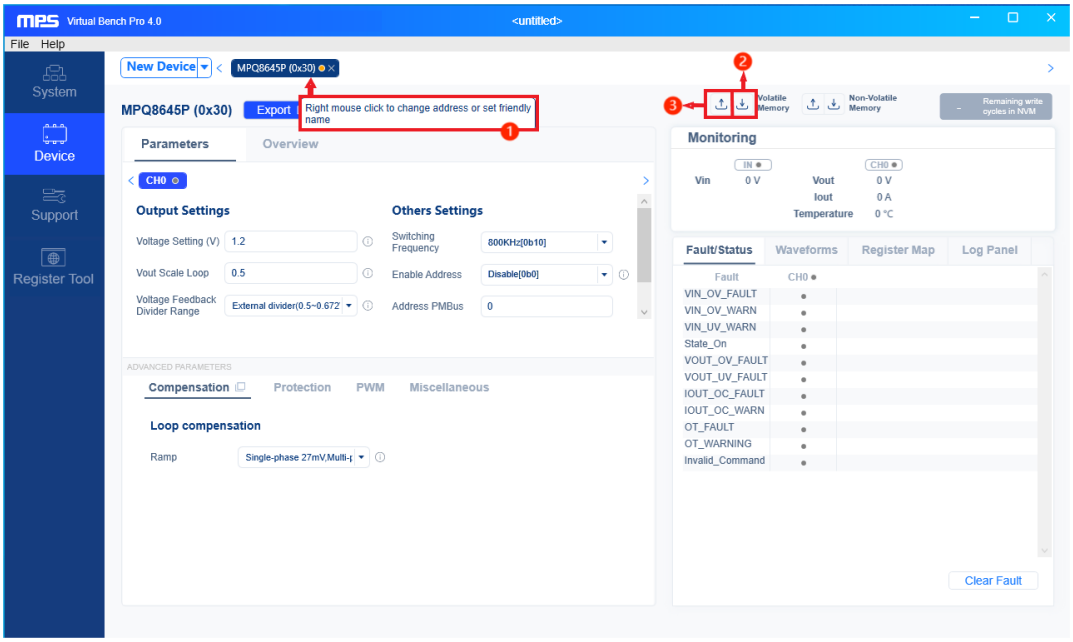
3.5 Import Local Configuration

Once a digital code is created, it allows users to load the digital code to VBP4.0. Select **File->Import ATE file**.

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After loading the digital code, the user can see the same configurations as the last saved.



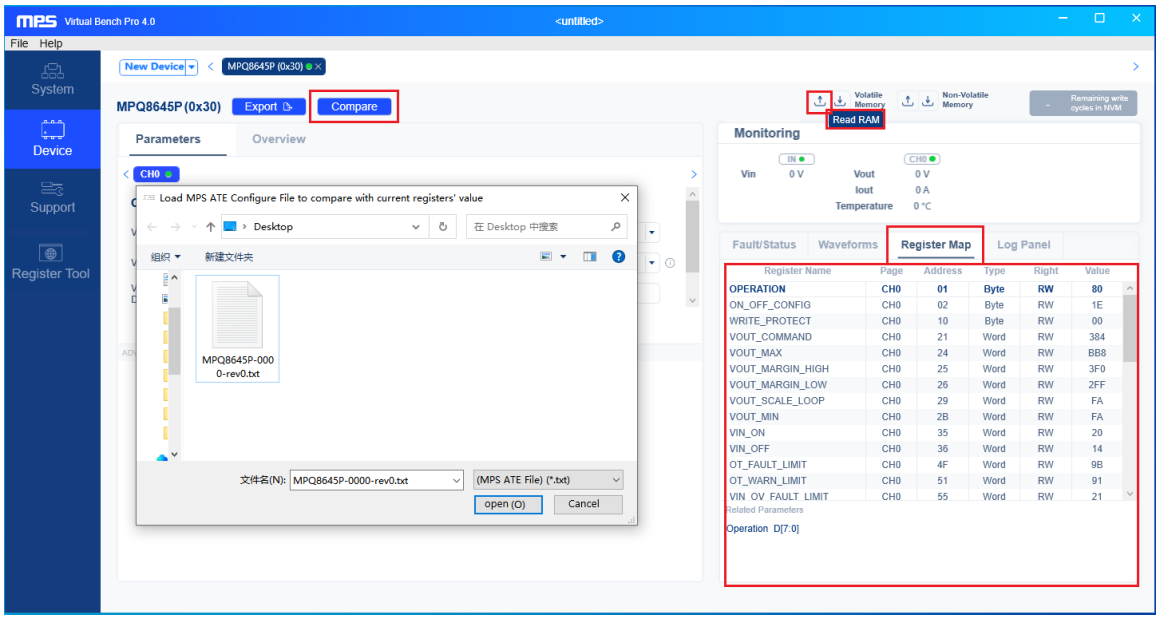
When the digital code is loaded successfully, the address of the chip may change and needs to be changed manually on the GUI (Note: How to change the address, please see step 3). After importing the digital code, if the address does not need to be changed, user also need to perform the following steps: need to click "**Write Ram**" first and then click "**Read Ram**", the purpose is to update the GUI parameters into the chip (**Note:** The value stored in the digital code may be different from the current value in the chip).

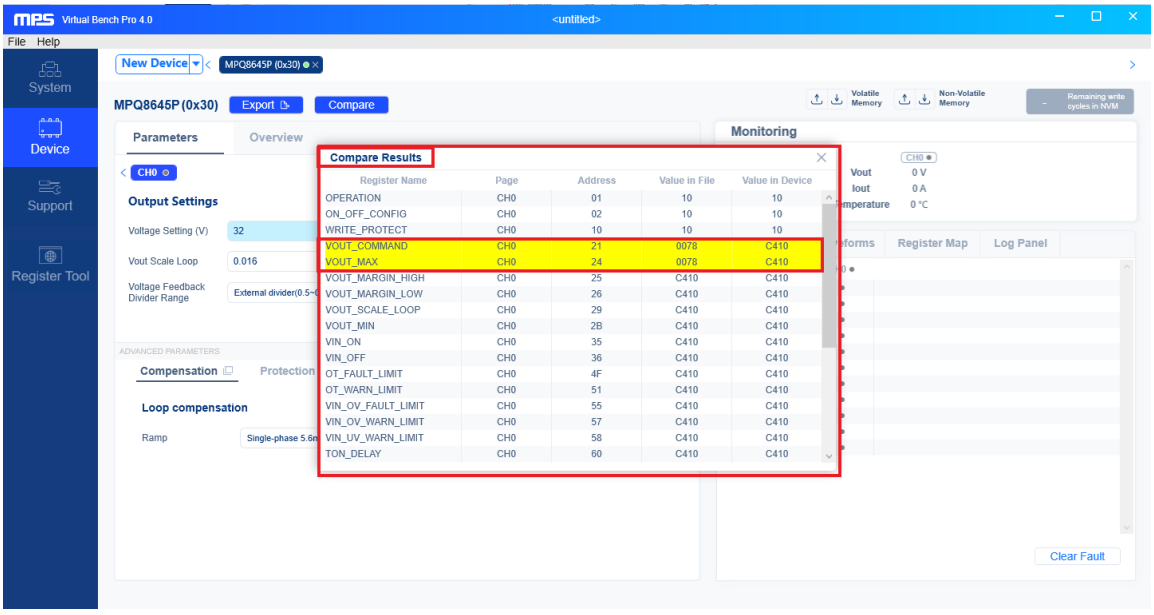
3.6 Compare the Configuration

In the Device panel, the **Compare** button is used to compare register values from ATE file with register values from device. Click **Compare** button, select ATE file, the window will pop up to display the

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result of the comparison (Note: the compare function will be disabled when the device is offline). If you disconnect the current device when opening it in the GUI, users will need to "**Read RAM**" after connecting the same device again. These operations are used to ensure that the current values are synchronized.



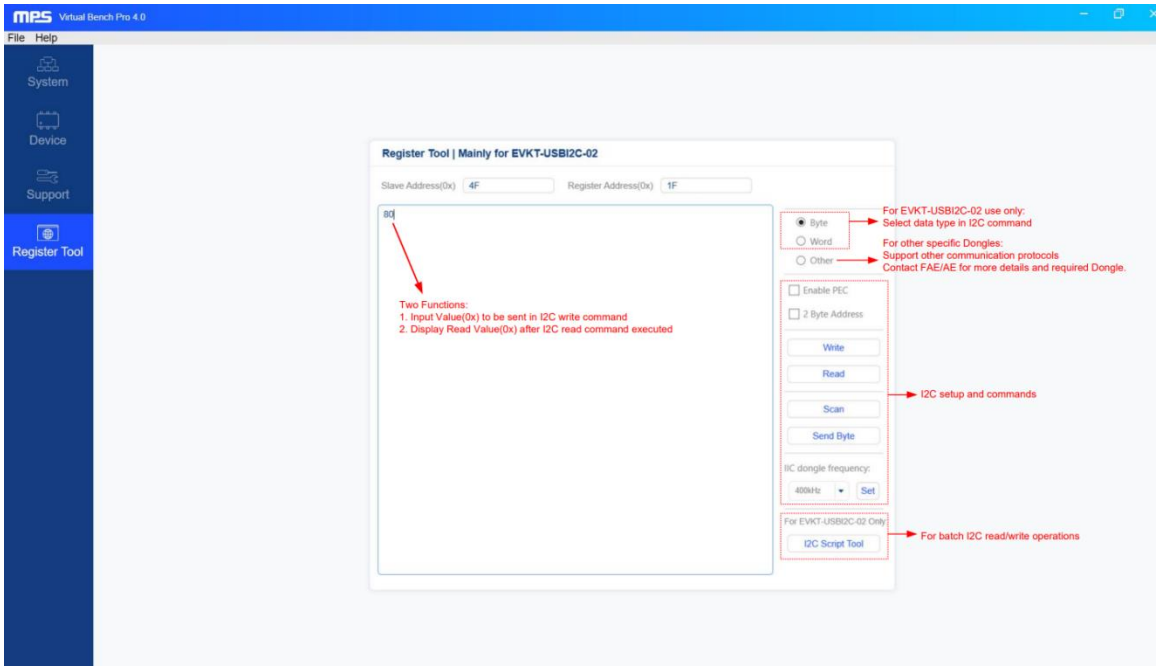


3.7 Register Tool

Additional operations for the connected part are supported in the Register Tool Panel. User should notice that Using Register Tool may affect the current chip configuration. So, proceed with caution.

Please refer to the screenshot below to get instructions and notice that when there is no connected device, the register tool will not work.

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The **I2C Script Tool** supports executing multiple I2C commands by sequence, includes:

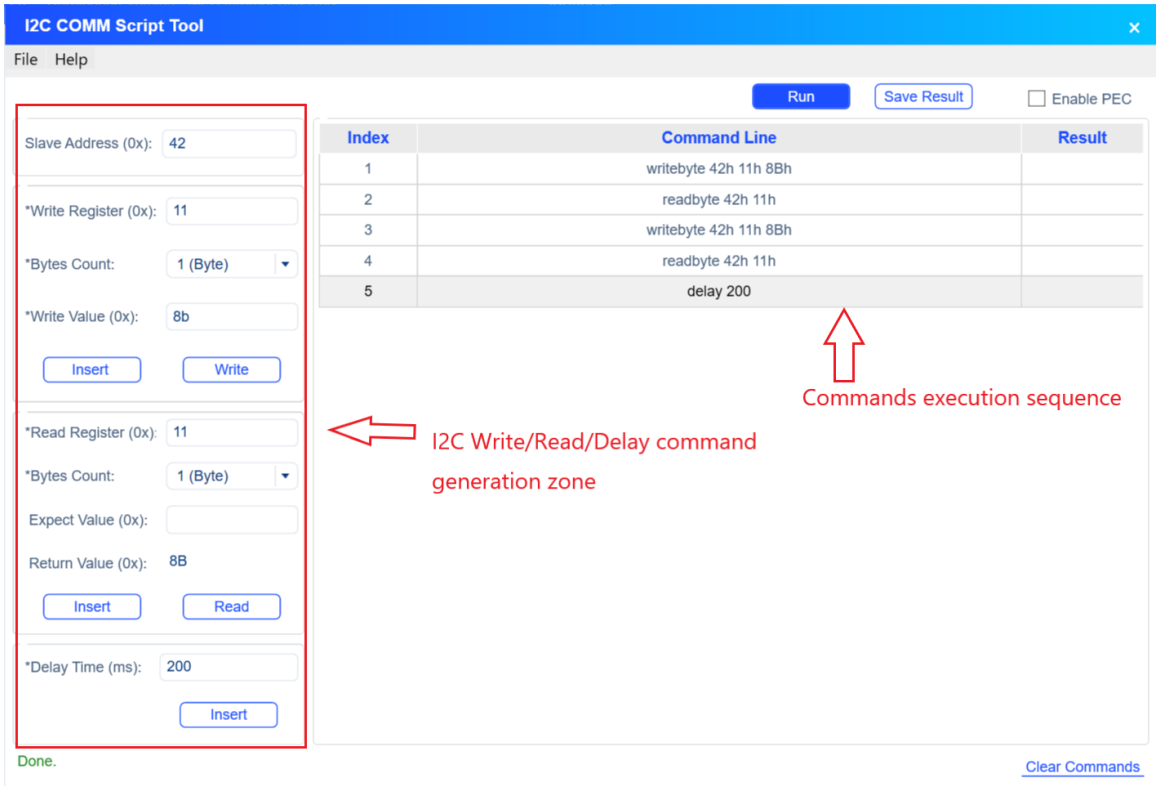
- **Command Generation Zone**: support I2C read/write and delay commands. User can input all required command information and click 'Insert' to add it to the right command list.
- **Commands Execution Sequence**: displays all added commands in sequence. Mouse right-clicking can delete the selected command.
- **Run**: click to execute all commands in the list.

  [Note]: If the read command has an expected value, the readback and expected value will be compared. If the results are inconsistent, the row will be marked red.
- **Clear Commands**: click to clear all commands in current list.
- **Save Result**: click to save all results to a local txt file.
- **File**:

  Load Script: click to load the local txt file into the tool.

  Save Script: click to save all scripts to a local txt file.
- **Enable PEC**: check if PEC is needed.
- **Help**: click to view the user manual.

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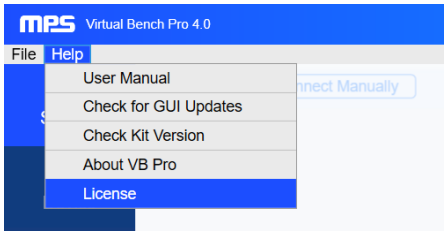
The commands supported by I2C Script Tool is shown as below:

Function	Sample	Details
read byte	***readbyte*** 01h D1h	***readbyte***: key word to read byte value ***readword***: key word to read word value 01h: slave address (hex) D1h: register address (hex) 00h/0000h: user expect value (use for verification)(hex)
	readbyte 01h D1h 00h	
read word	***readword*** 01h D1h	
	readword 01h D1h 0000h	
write byte	***writebyte*** 01h D1h EFh	***writebyte***: key word to write byte value ***writeword***: key word to write word value 01h: slave address (hex) D1h: register address (hex) EFh: byte value need to be written 01EFh: word value need to be written
write word	***writeword*** 01h D1h 01EFh	
send	***send*** 01h D1h	send command
delay	***delay*** 50	delay 50 ms
loop	***loop 3*** ***readbyte*** 01h D1h 00h ***writebyte*** 01h D1h ***endloop***	***loop-endloop***: key word to achieve a loop 3: loop time (Support nesting)

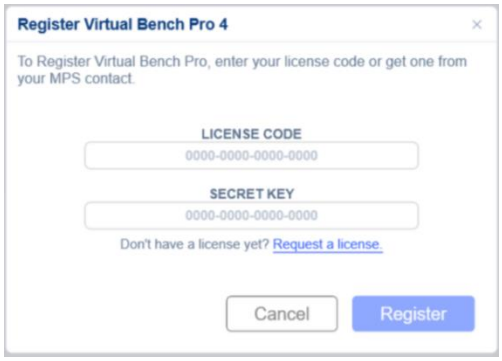
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4 License

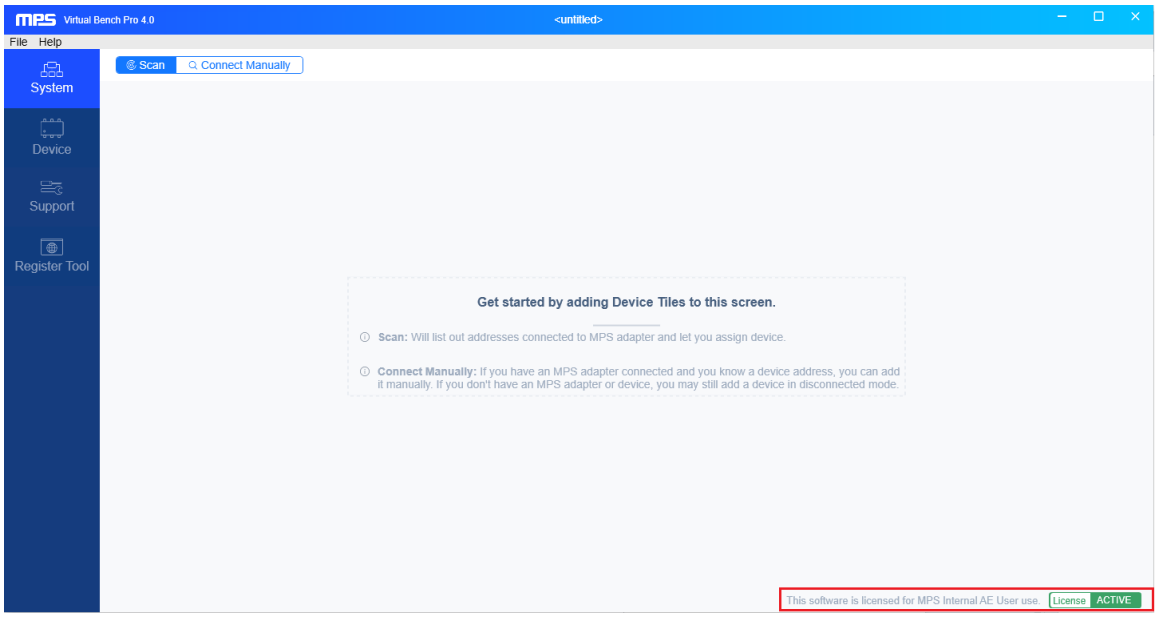
VBP4.0 users don't need a license to configure all the devices published in VBP4.0. In case a user needs to configure an unpublished device that is supported by VBP4.0, the user may talk to MPS Application Engineer in charge and obtain a customer license form MPS. Click **Help->License** menu.



Follow the Request a license link to submit a license request. Upon approval, a license will be emailed to the requester.

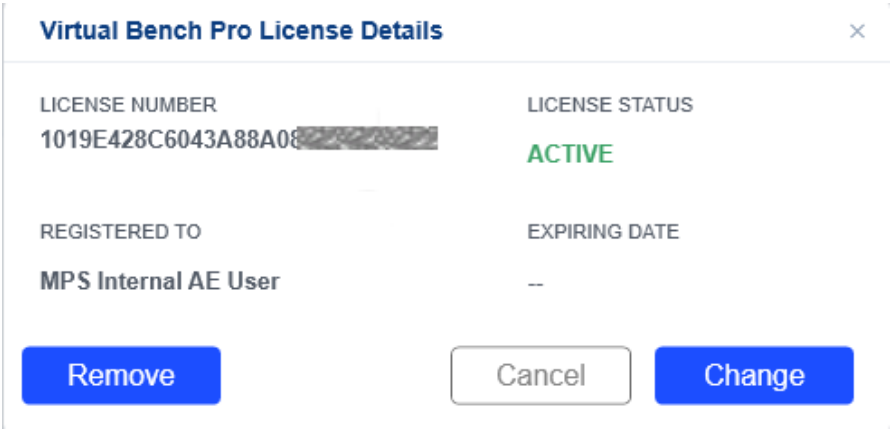


The requester needs to fill in the LICENSE CODE and SECRET KEY from the email, then click **Register** button. The license information will be displayed at the bottom right corner. After the license is successfully registered, the product will be updated.



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Click **Help->License** menu to see detail information of license.



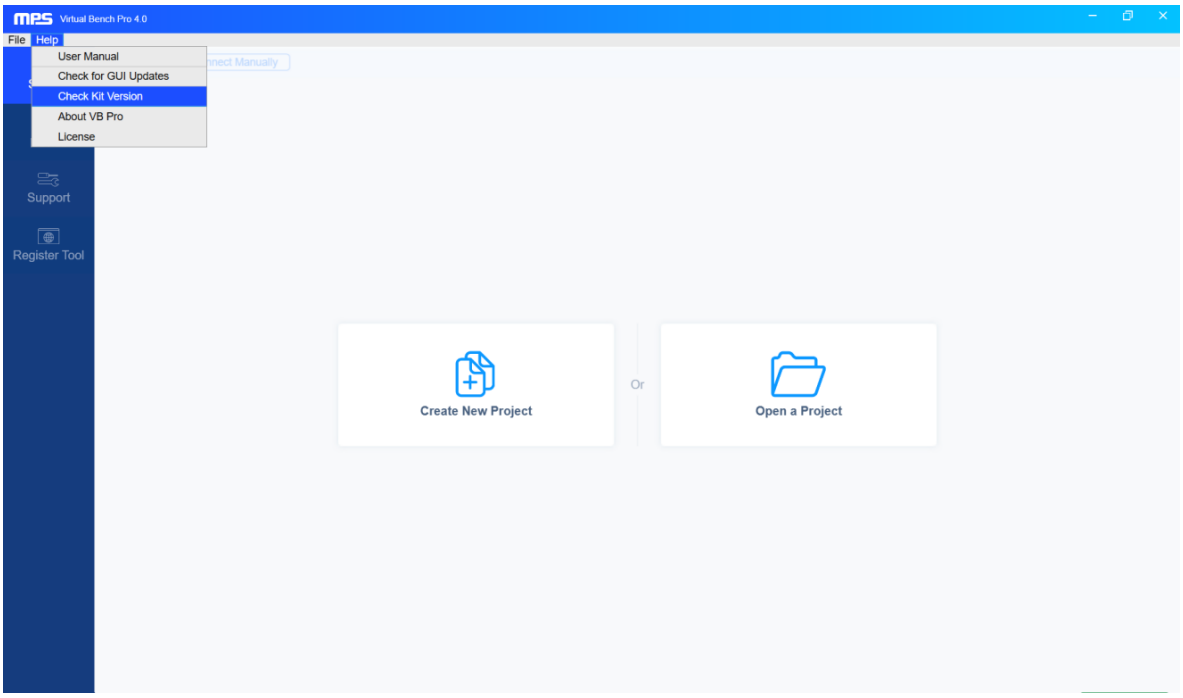
In the License details window user also allows to update license information. Click **Change** button, users can change license information such as switching user account. Click **Remove** button, the current license will be deleted.

Note : the opened parts need to be closed before performing the operations.

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5 Dongle Kit Firmware

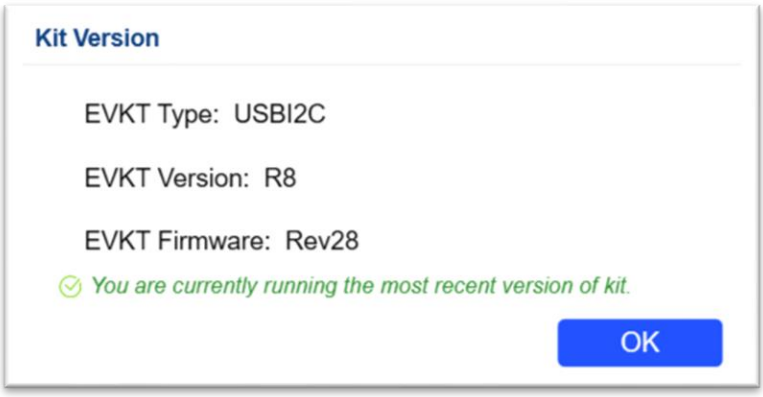
Click “Help->Check Kit Version” to see firmware version of the dongle and check if it needs to update.



5.1 Kit Version

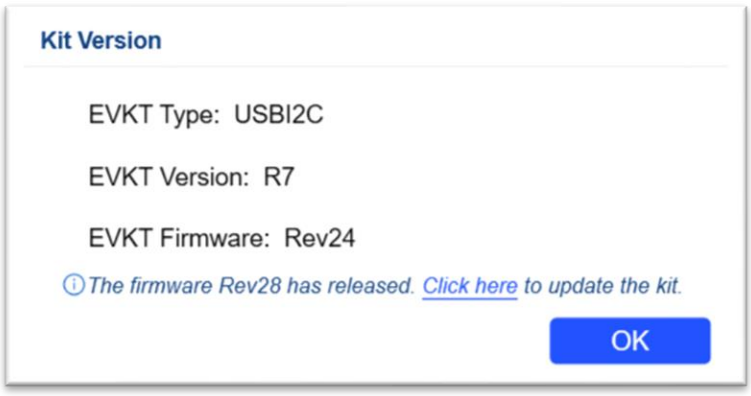
It will take a while to read the kit version, and the information will pop up when it’s done. Only EVKT-USBI2C-02 and EVKT-AUTOMOTIVE-00A support for firmware version checking.

- The kit firmware version is up to date:

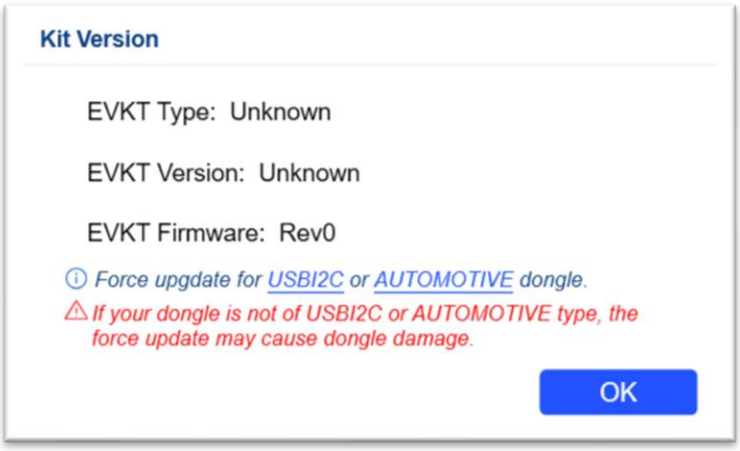


- The kit firmware version is outdated and supports updating:

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- Failed to read the kit version:



5.2 Kit Update

If the dongle firmware is not the latest version, VBP4.0 supports to upgrade the dongle firmware. Please note that **only EVKT-USBI2C-02 (⩾Rev24) and EVKT-AUTOMOTIVE-00A support firmware update function**.

Hit "*Click here*" to update the kit.



When the upgrade is done, the latest information will be displayed.

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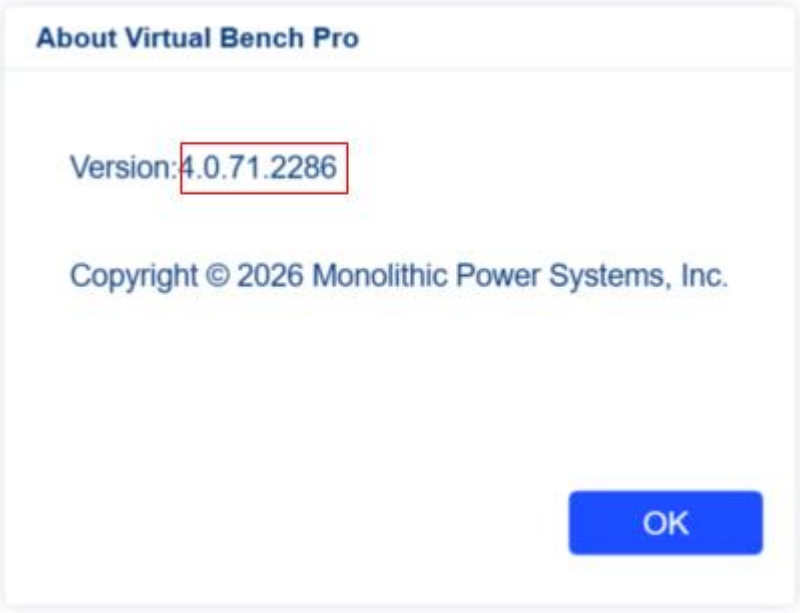


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6 GUI Version/Update

6.1 GUI Version

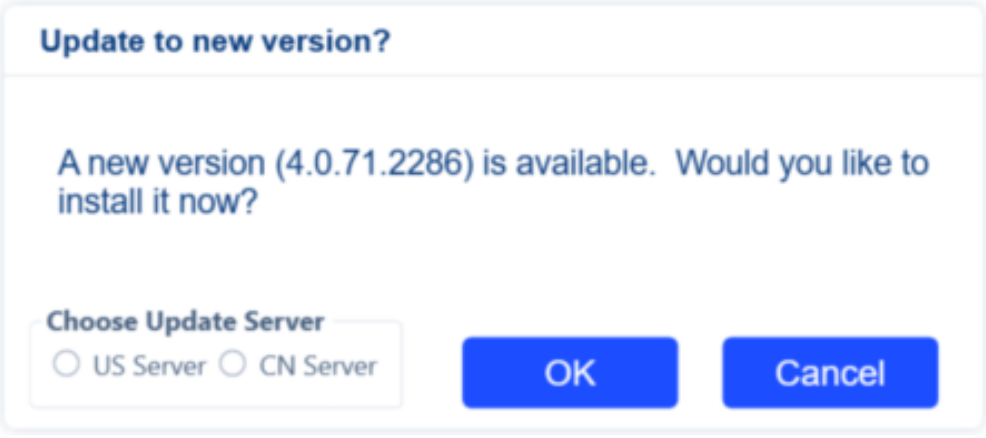
Click “**Help->About VB Pro**” to check GUI version.



6.2 GUI Update

The GUI offers two methods for updates:

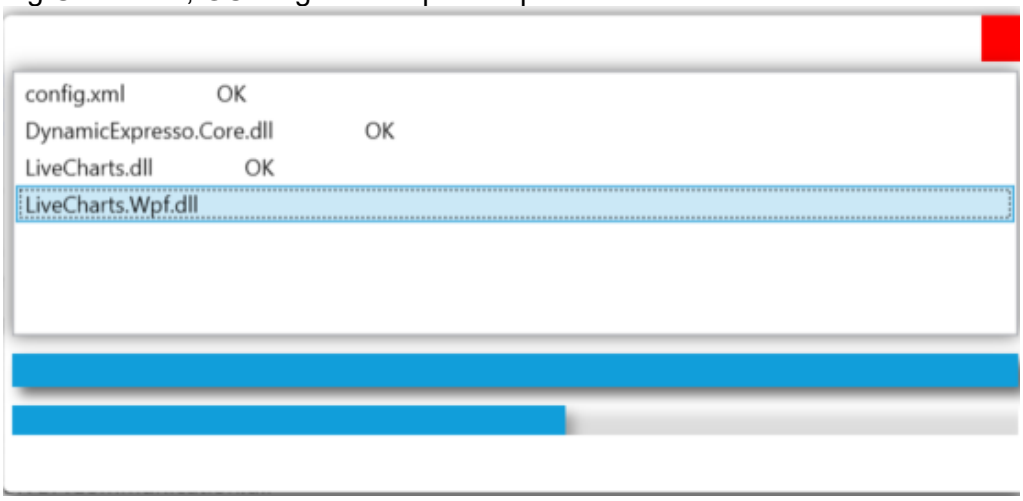
1. When launching GUI, an update prompt box will pop up if the currently used version is not the latest one. Choose update server based on the user's location before clicking the **OK** button.



(Note: CN server for users in China, US server for all other regions. The selection of the update server will affect the update duration. Please make selection carefully.)

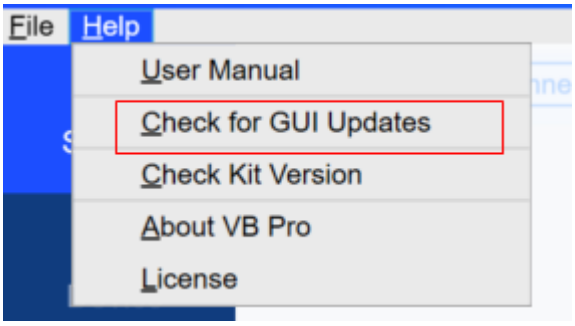
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After clicking **OK** button, GUI begins the update operation.

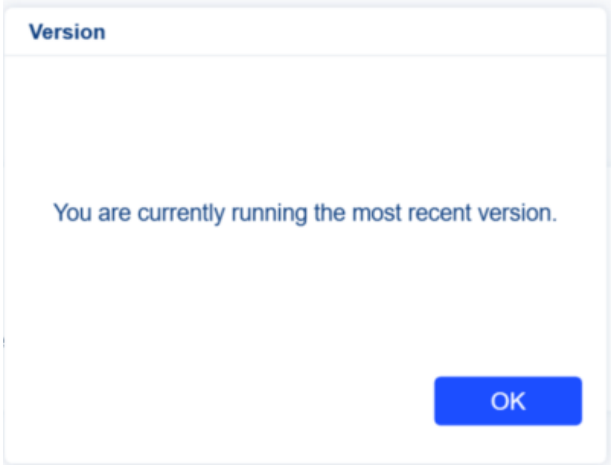


(Note: Do not perform any other operations before the update is completed to avoid failure.)

2. Click **Help > Check for GUI Updates** to update the GUI version when required. The procedure is identical to Method One.



If the version is the latest, the following window will pop up.



After the update is completed, refer to Section 6.1 for GUI version verification

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7 Troubleshooting

If the user can't find any device after environment setup, user may do the following steps to check the issues:

1) Make sure the EVB works well. (For example, measure the output voltage on EVB with multimeter)

2) If there are other devices connected with USB on PC except MPS dongle, remove them and try again.

3) Click Help->Check Kit Version, check if MPS dongle works.

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8 Contacts for support

Users can always go to MPS website https://www.monolithicpower.com/en/contact.html for help and support.

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